HISTOLOGY OF BLOOD VESSELS

Learning Objectives

At the end of the lecture, student should be able to;

- Identify the 3 layers found in blood vessels.
- Distinguish between arteries ,veins and capillaries.
- Identify and differentiate b/w different types of veins.
- Identify and differentiate b/w different types of arteries.
- Identify and differentiate b/w different types of capillaries.

Blood Circulatory System

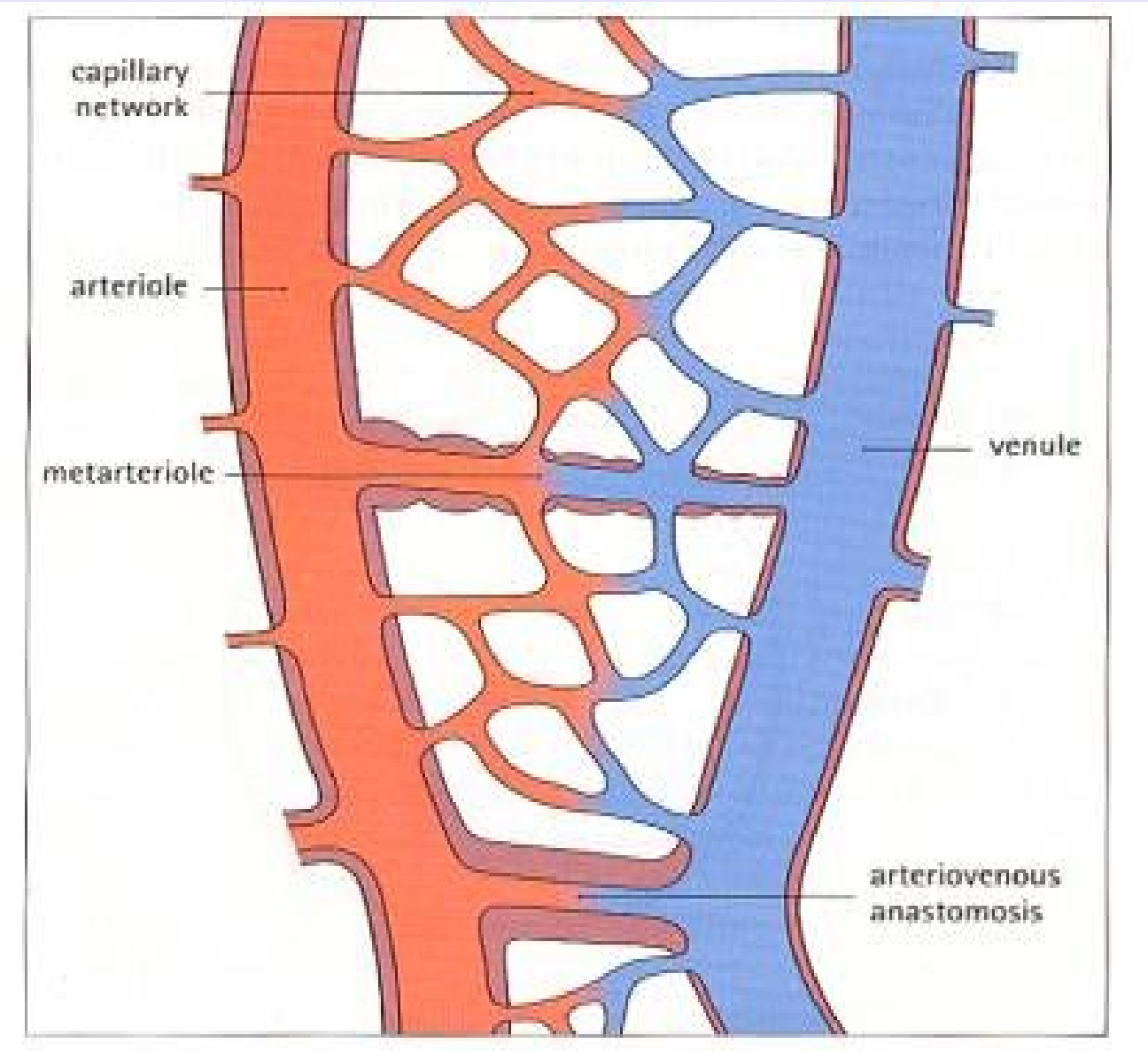
- Comprises of a central muscular pump, the heart.
- and a circuit of vessels through which the blood flows.
- The blood circulatory system has three kinds of vessels

arteries

veins

capillaries

Arteries Veins Capillaries



Circulatory System-Layout

- The whole circulatory system has a common basic structure :
- Composed of three layers
- **Tunica intima**
- **Tunica media**
- **Tunica adventitia**

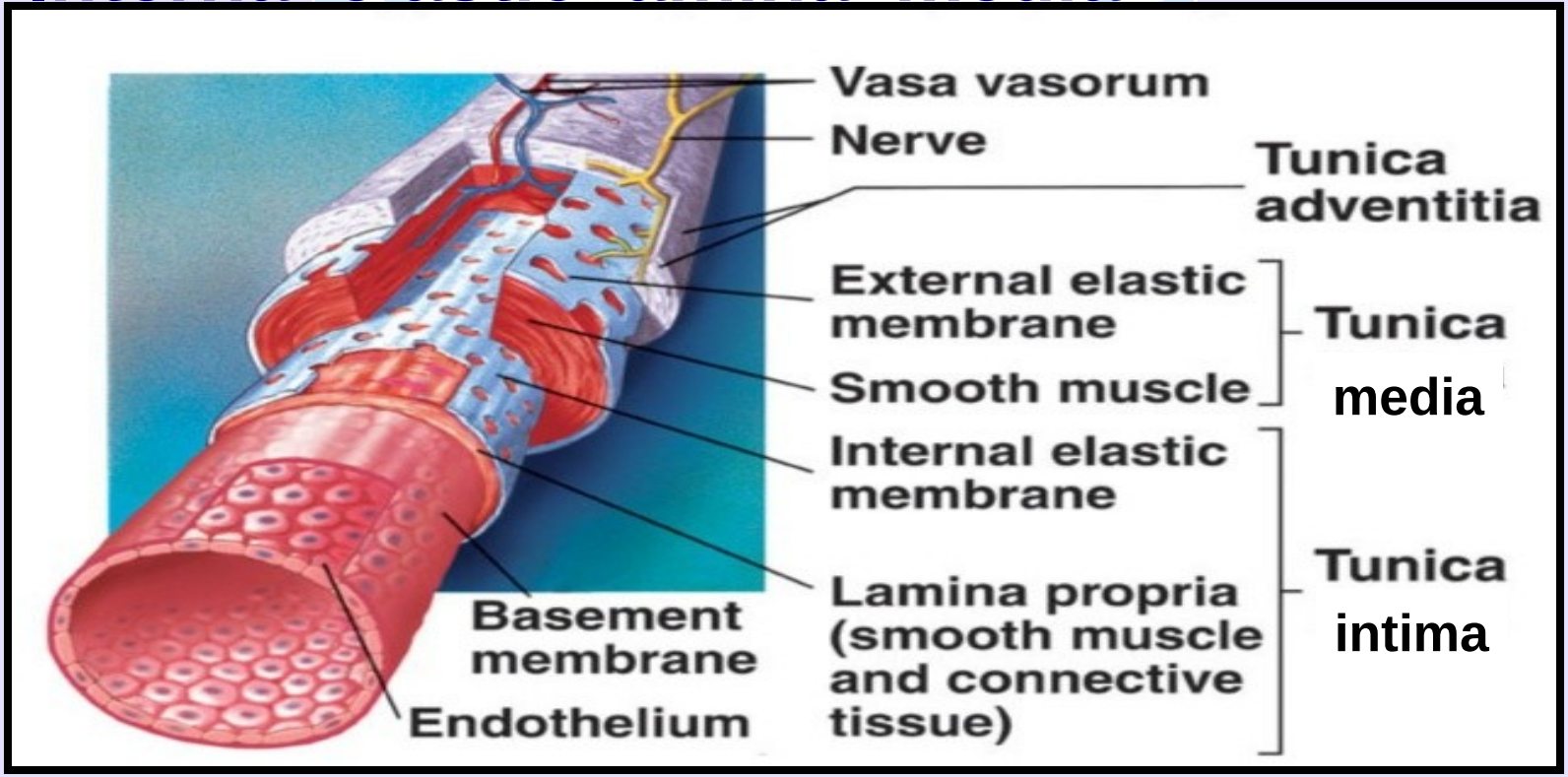
Tunica intima

Innermost and consists of:-

a)**Endothelium.** Simple squamous epithelium

b) **Subendothelium** A thin layer of loose CT (connective tissue)

c) **Internal elastic lamina media**

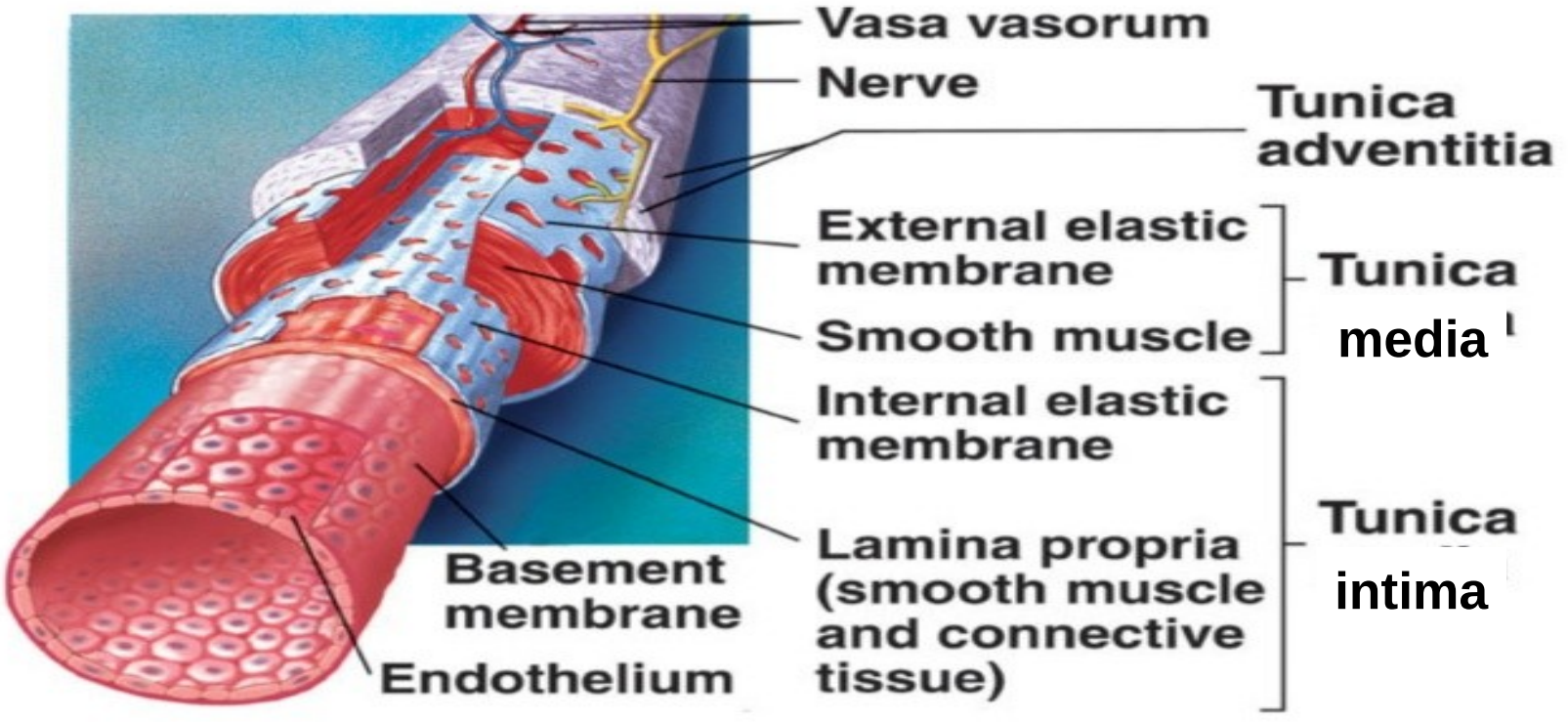


Tunica media

- **Middle layer**

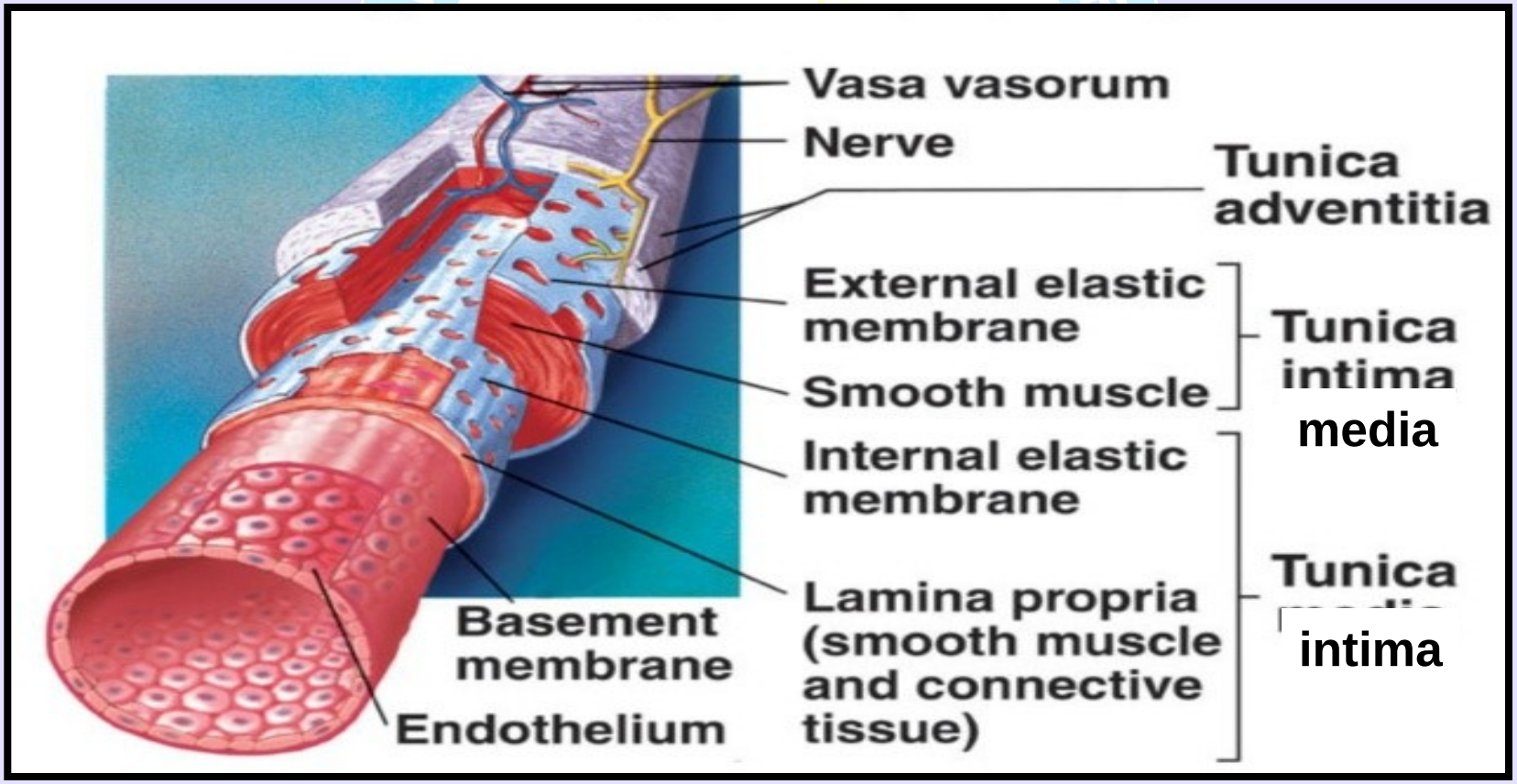
a) Concentric layers of smooth muscle cells

b) External elastic lamina

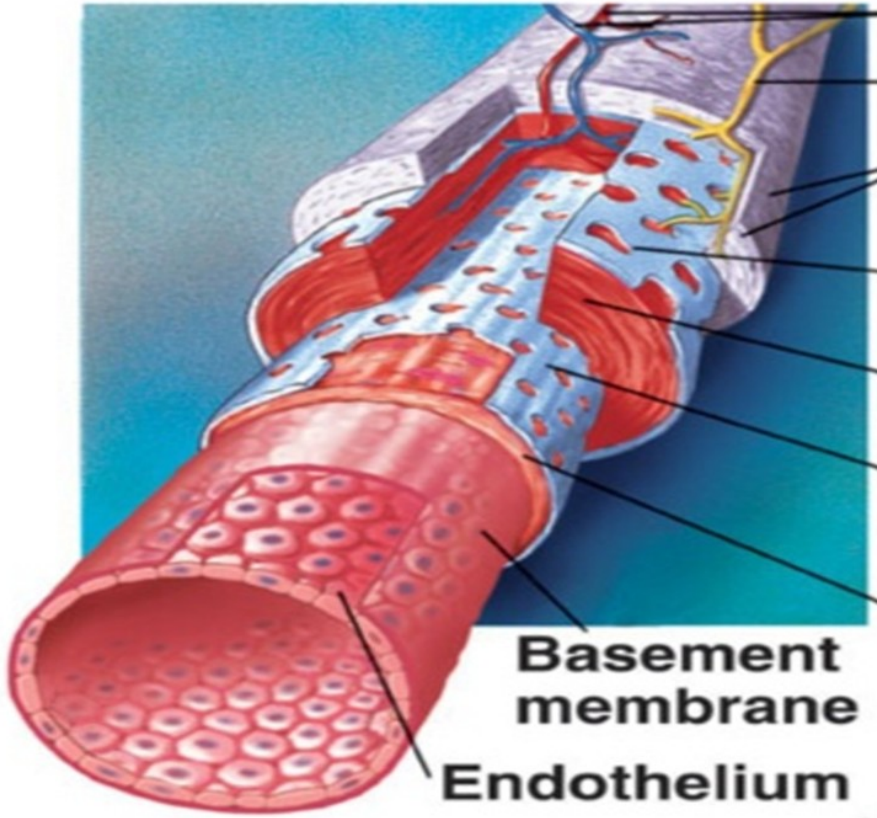


Tunica Adventitia

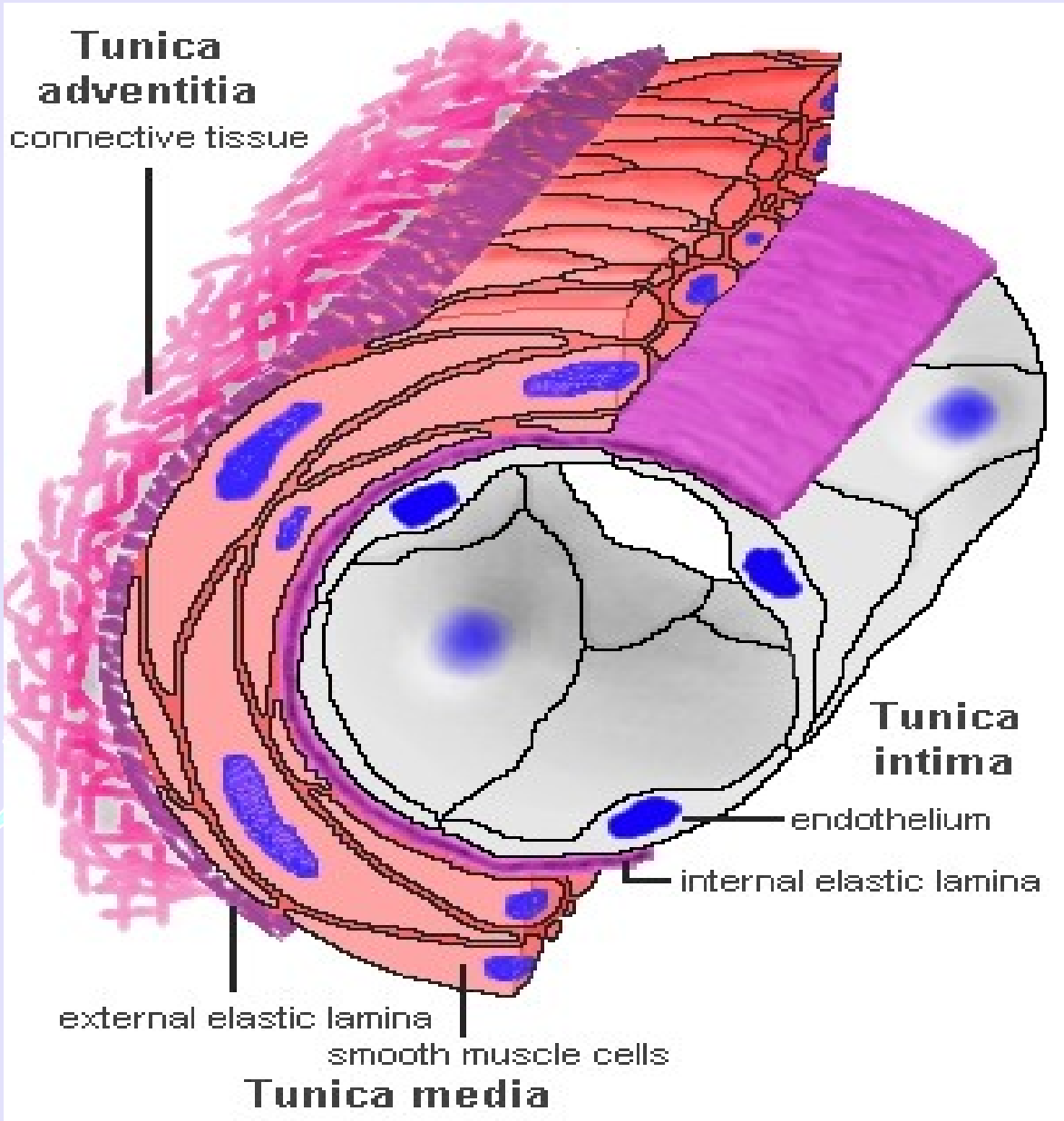
- Outermost layer
- Composed of supporting tissue



Basic structure of the vessel

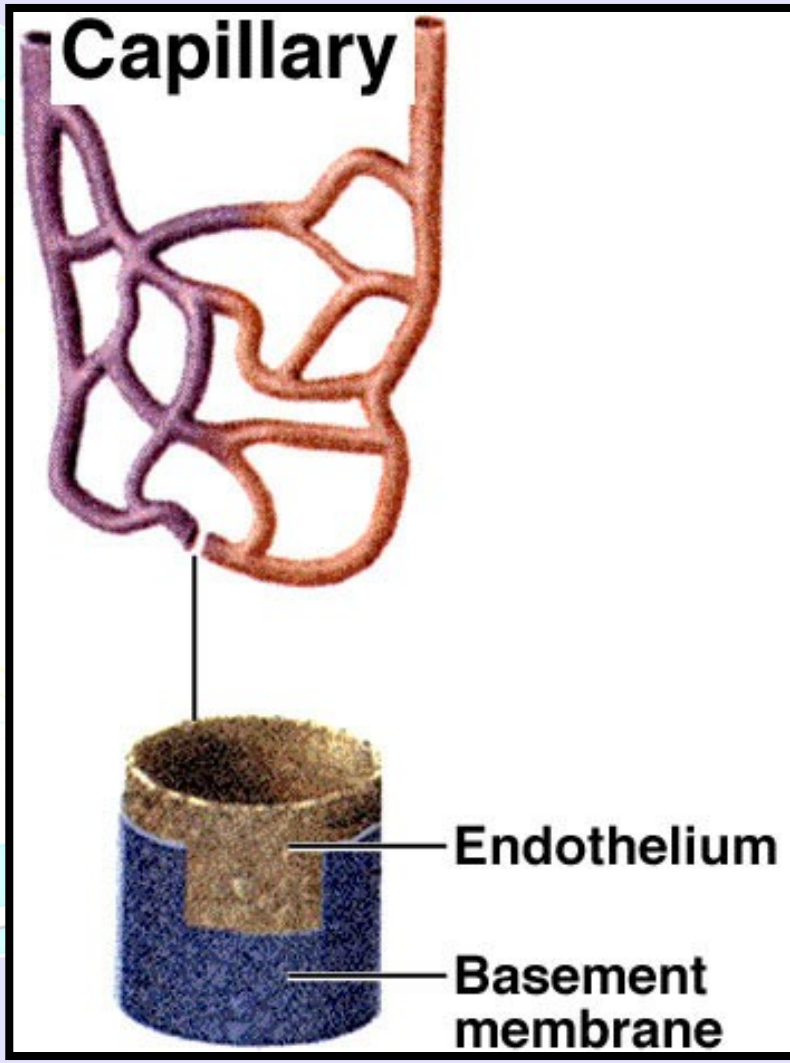


Structure of vessel walls



CAPILLARIES

- Endothelial tubes.
- Diameter – average 7 to 9 µm
- Length – usually 0.25 to 1mm.
- Usually arranged in network form.
- Endothelial cells- polygonal in shape & arranged as a mosaic.
- Cell borders are wavy or serrated, and held together by tight junctions.
- Centrally located nucleus bulge into the lumen of capillary.



CAPILLARIES

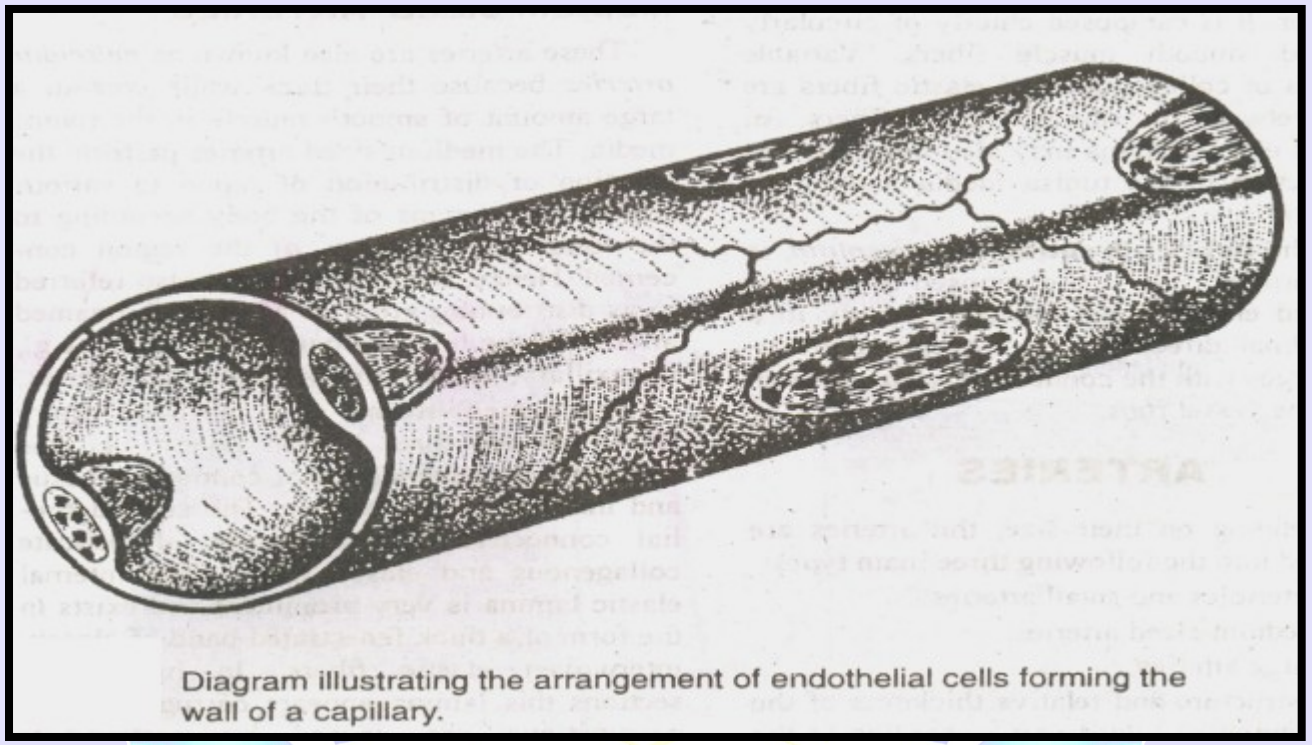


Diagram illustrating the arrangement of endothelial cells forming the wall of a capillary.

Endothelial cells usually rest on basal lamina.

Outer to basal lamina, there is thin layer of CT consists of delicate collagenous & reticular fibers, and pericytes.

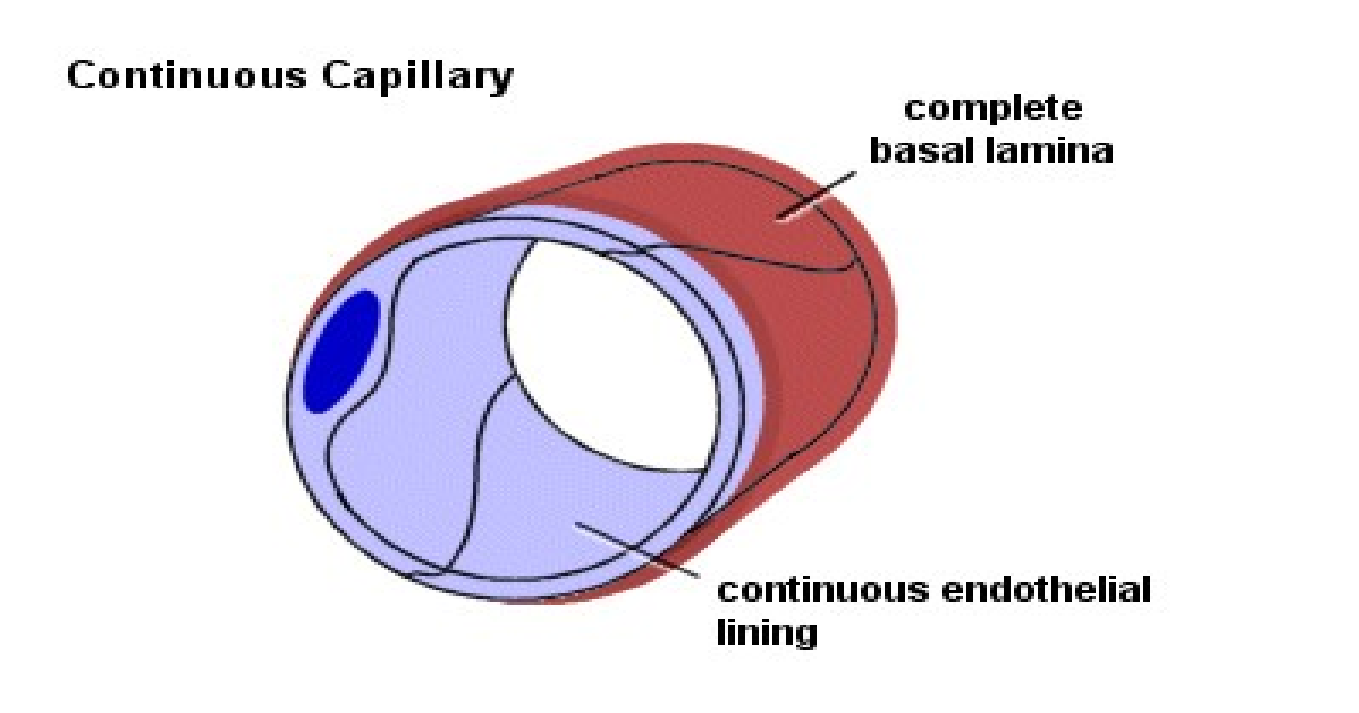
Classification of Capillaries

3 types, according to the structure of endothelial wall.

1. **Continuous or Somatic.**
2. **Fenestrated or Visceral.**
3. **Sinusoidal or Sinusoid.**

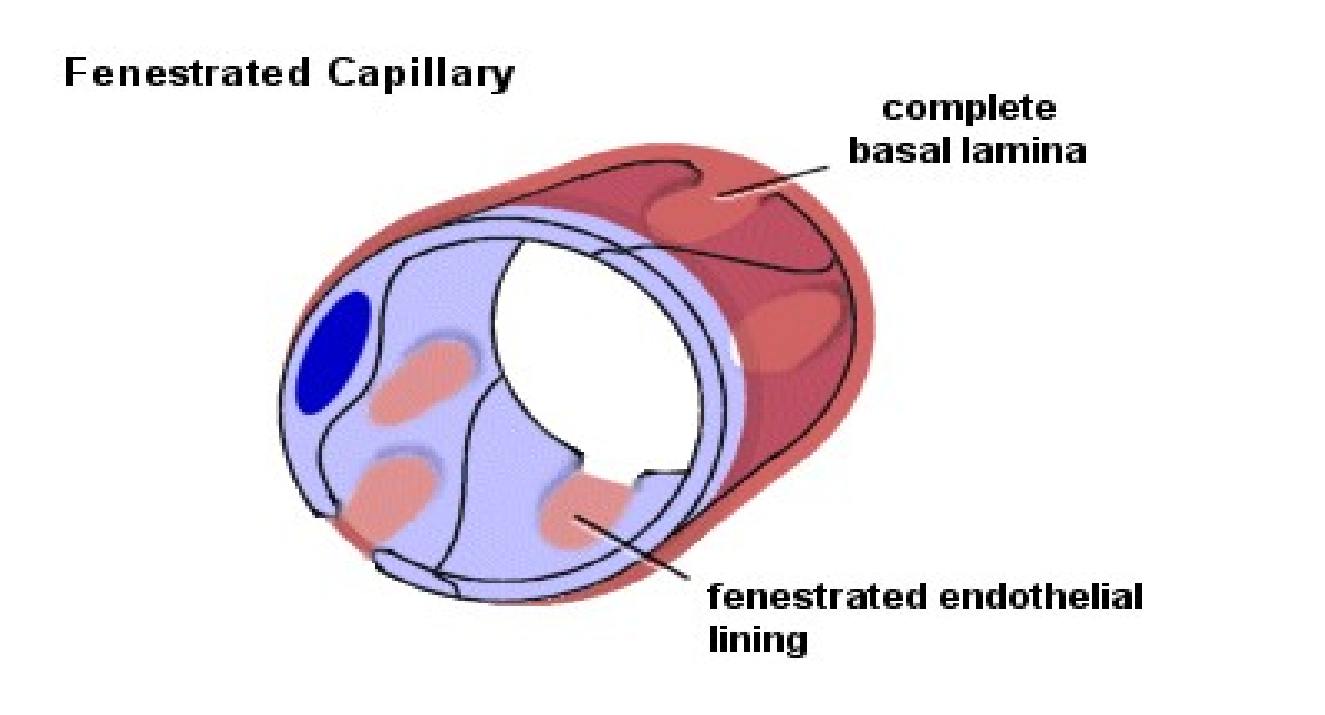
Continuous Capillaries

- No pores, gapes and discontinuities in their walls.
- Usually numerous pinocytotic vesicles in the cytoplasm of cells for transport of fluid across the capillary wall.

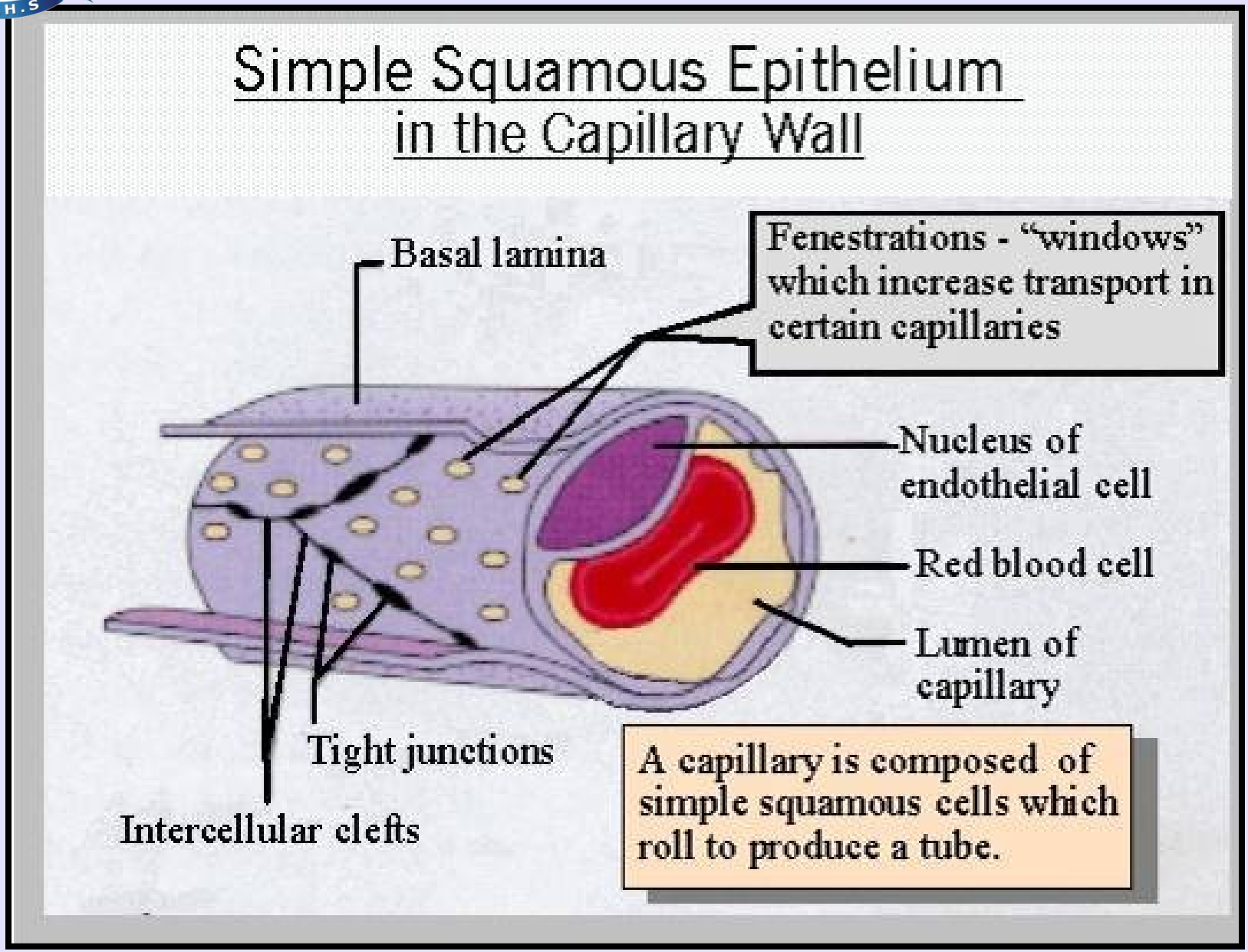


Fenestrated Capillaries

- Numerous circular pores or fenestrae (60-80 nm) in their wall.
- Pores are closed by diaphragm (except glomerular capillaries).
- A continuous basal lamina is present.

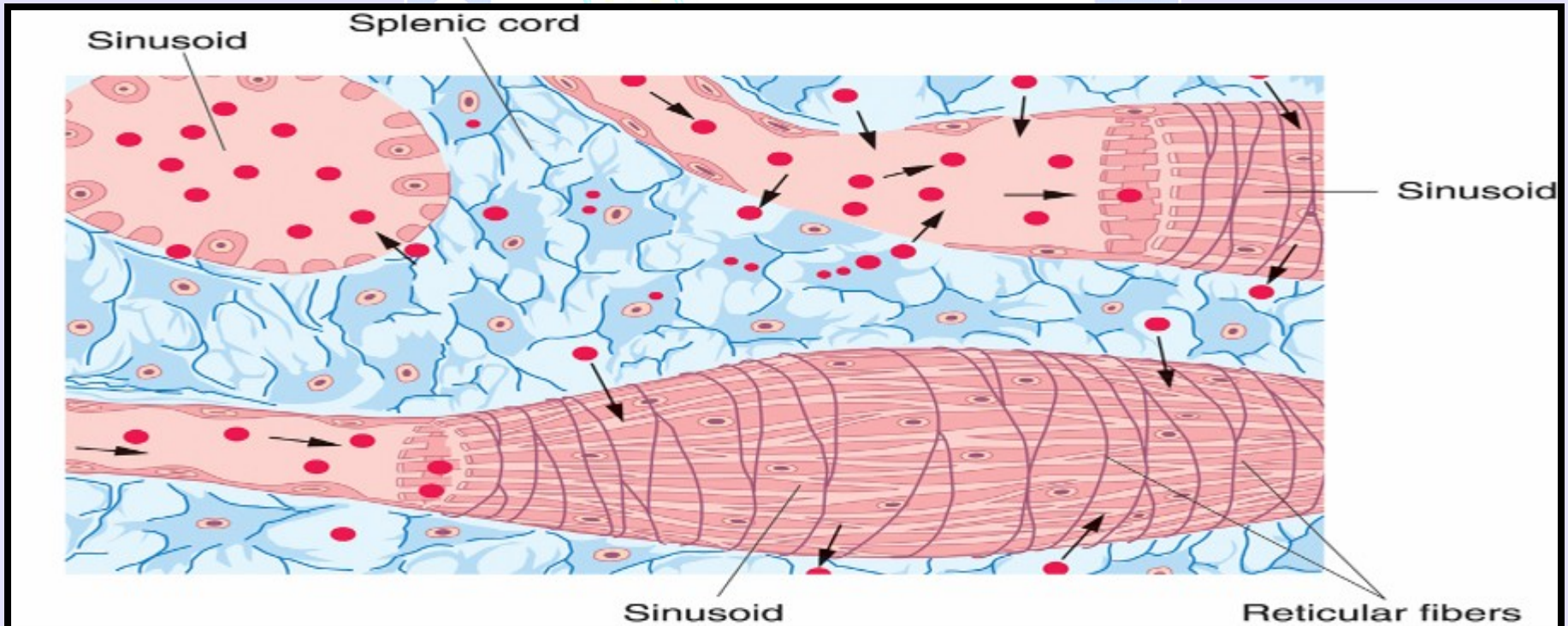


Fenestrated capillaries



Sinusoidal Capillaries

- Enlarged diameter(30-40 um). Irregular & tortuous path.
- Wide gaps b/w cells.
- Multiple pores (no diaphragm) in cell wall
- Incomplete basal lamina.



ARTERIES

According to size, 4 types:-

1. **Arterioles**
2. **Small Arteries**
3. **Medium size Arteries**
4. **Large Arteries**

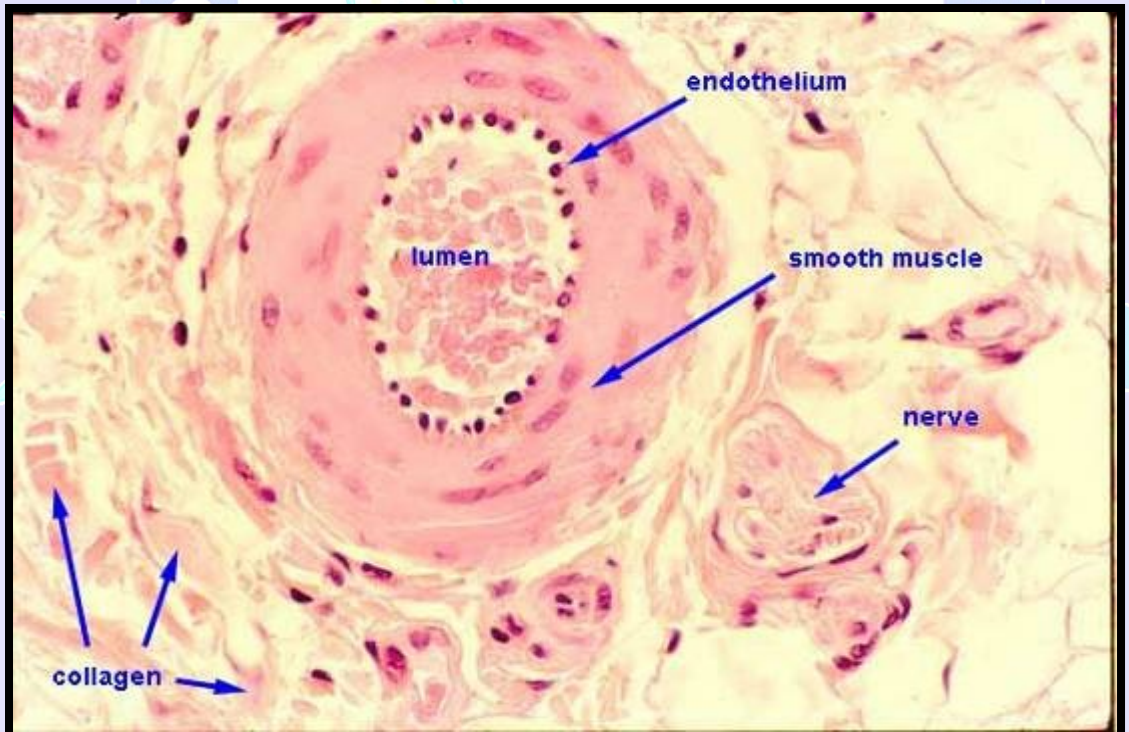
ARTERIOLES

Less than 0.5 mm diameter (narrow lumen &thick wall).

T. Intima : Only Endothelium

T. media: 1 or 2 layers

T. Adventitia: Thin, but Relatively thick.



SMALL ARERTIES

T. intima: Endothelium

Very thin Sub – endothelial tissue

T. media: up to 8 layers

T. Adventitia: Thin

MEDIUM SIZE ARTERIES

(Muscular artery OR Distributing Artery)
Examples: Named Arteries.

T. intima:

Endothelium:

Sub-endothelium: Somewhat thicker, few smooth muscle fibers may be present.

Internal elastic lamina: Very prominent, Forms a thick fenestrated band of closely interwoven elastic fibers.

T. media:

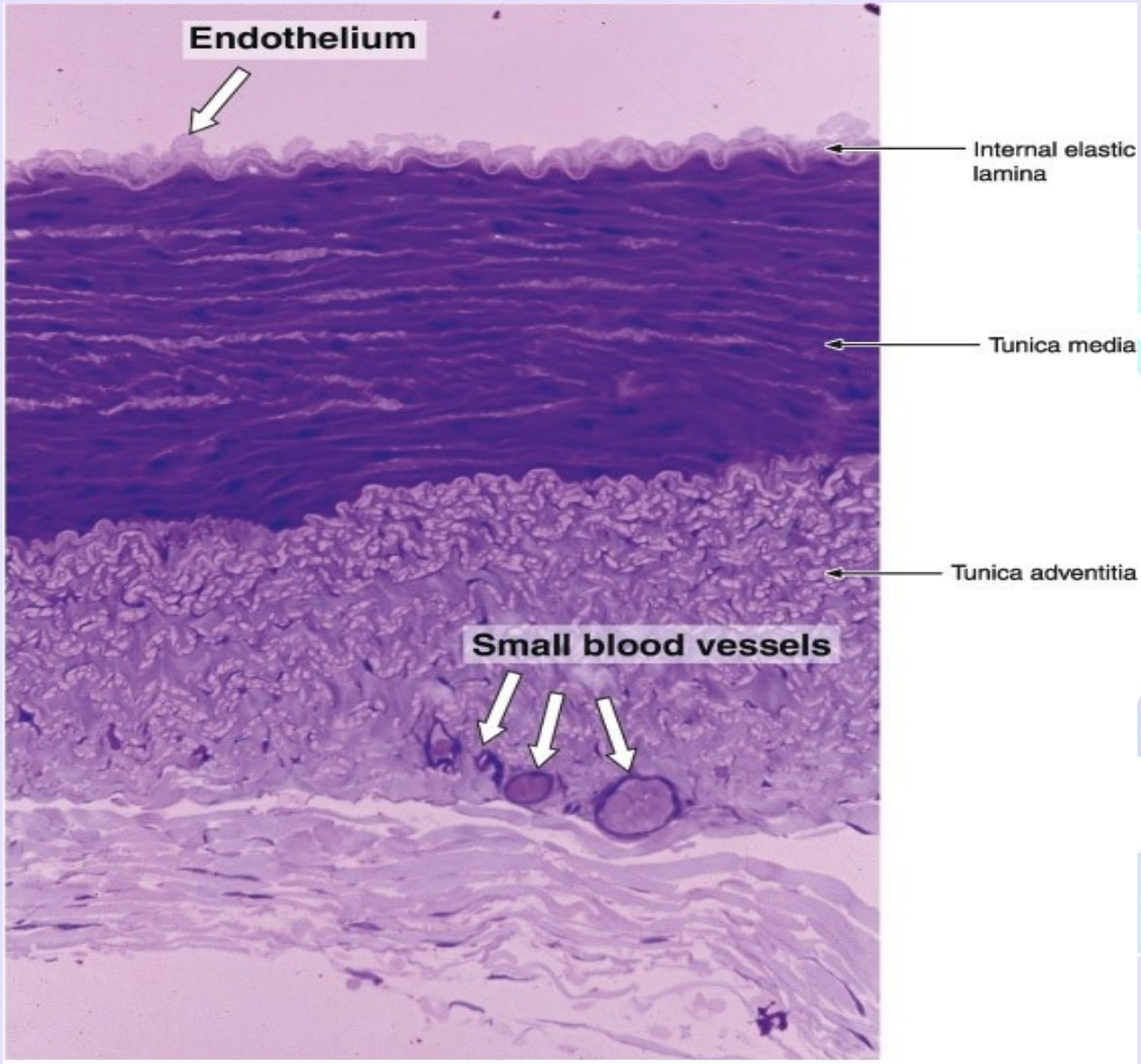
Several layers (up to 40), Small quantity of C.T. composed mainly of collagenous fibers.

External elastic lamina: May be seen

T. Adventitia:

Sufficiently thick (may be as thick as Tunica media).

muscular (medium caliber) artery

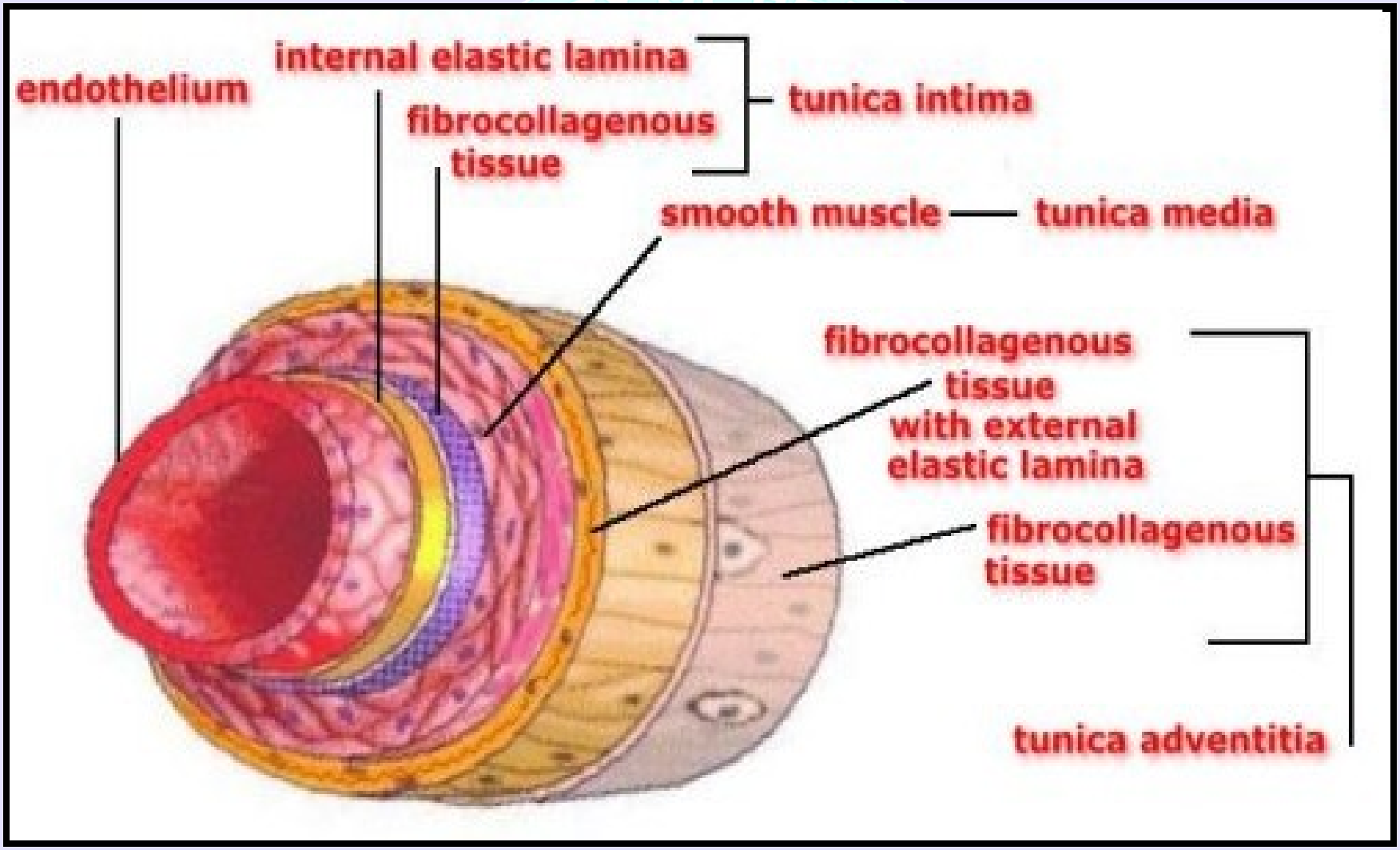


Source: Junqueira LC, Carneiro J: *Basic Histology: Text and Atlas*, 11th Edition: http://www.accessmedicine.com

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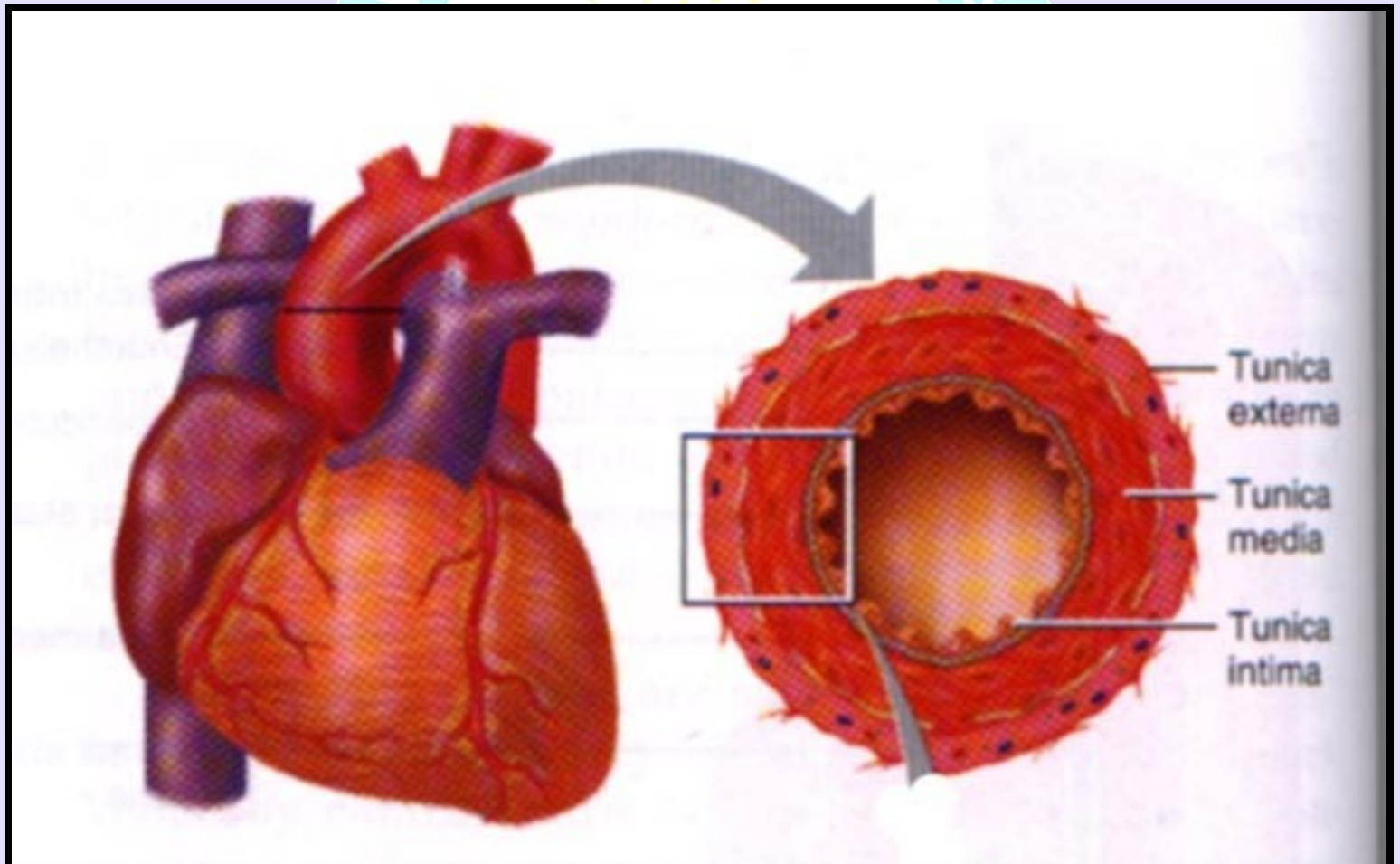
Transverse section showing part of a muscular (medium caliber) artery. Small blood vessels (vasa vasorum) are observed in the tunica adventitia

Muscular Artery; diagrammatic representation



LARGE ARTERIES

Elastic Artery (yellow) OR Conducting artery.
Examples: Aorta & Pulmonary trunk.



LARGE ARTERIES

T. intima:

Endothelium:

Sub-endothelium: Thicker, In addition to Collegen & Elastic fibers, fibroblasts and few smooth muscle fibers.

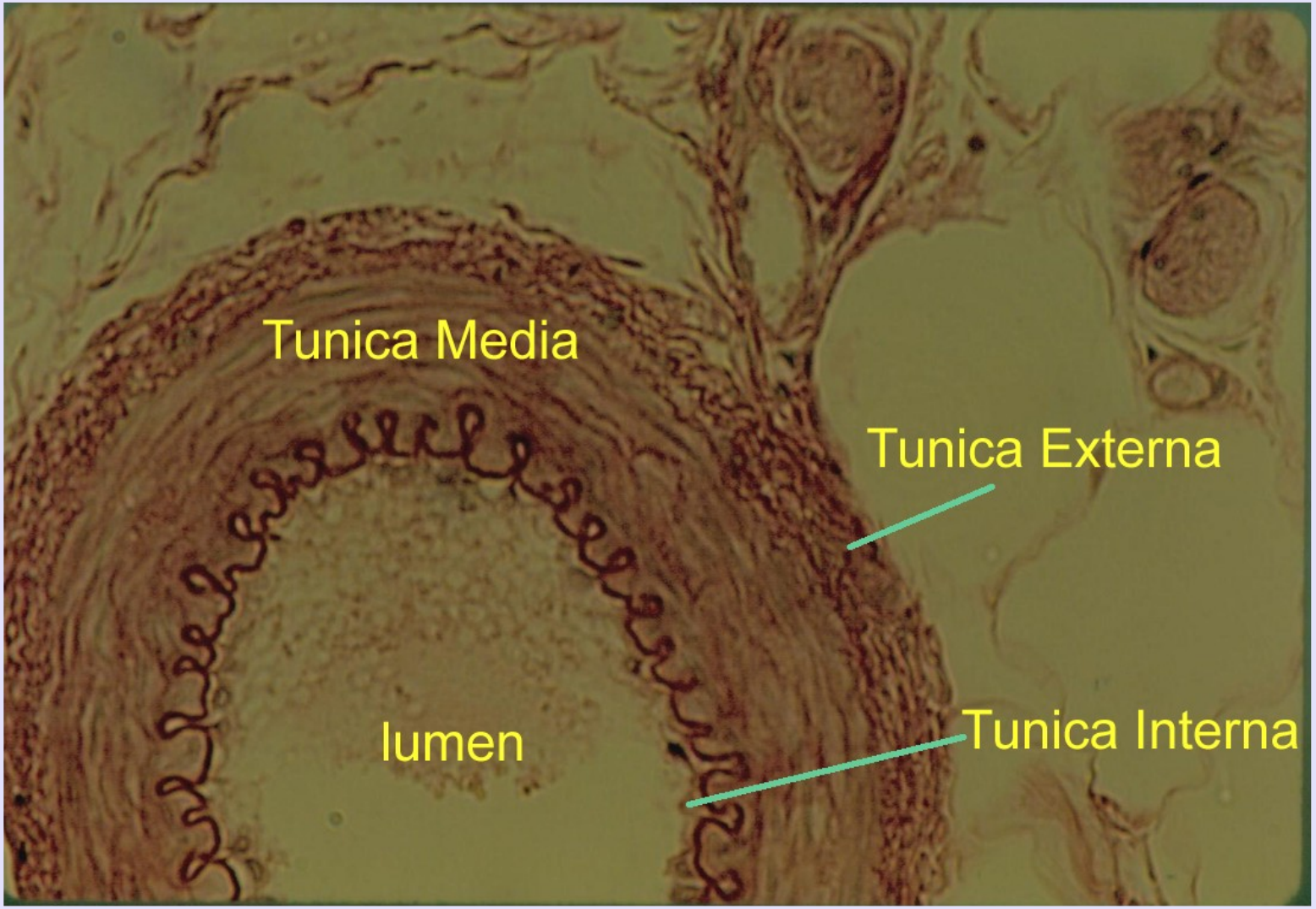
Internal elastic lamina: Not separately Identified

T. media: A series of concentrically arranged fenestrated elastic membranes. Number increased with age (40 in new born to 70 in adult).

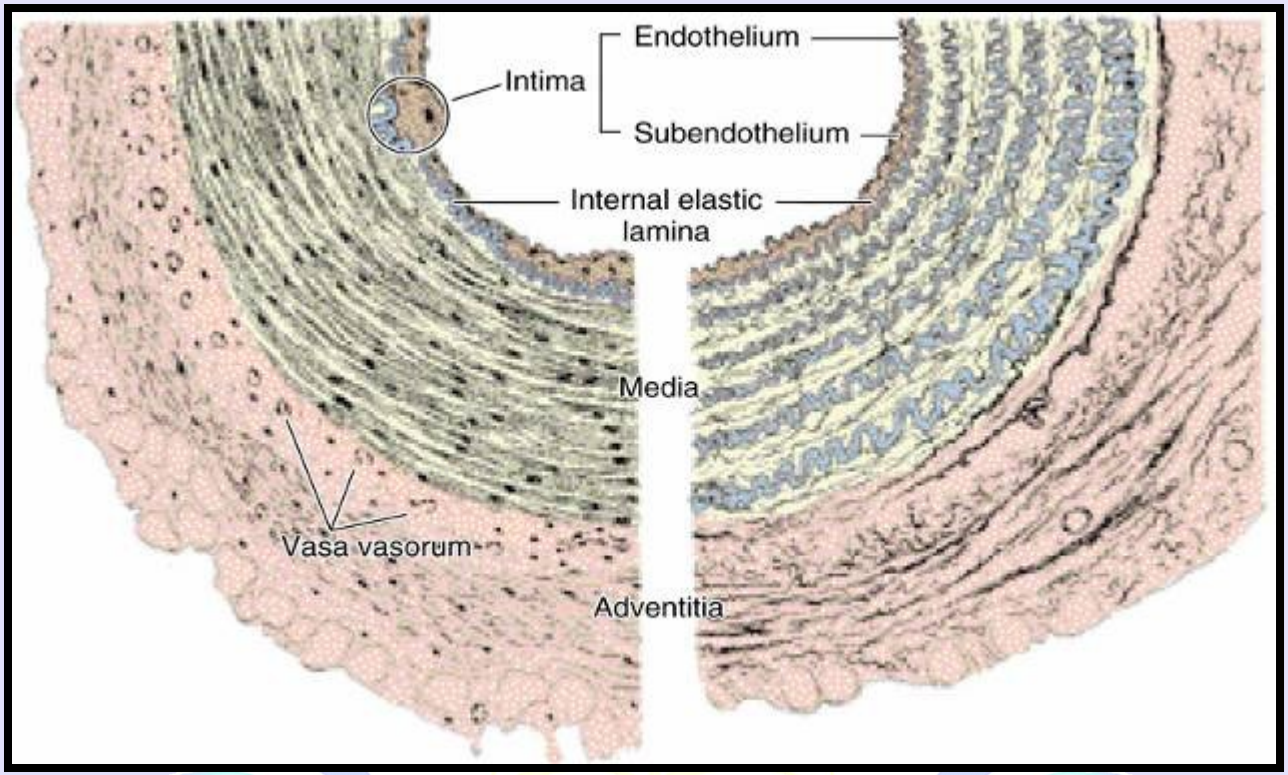
External elastic lamina: Absent.

T. Adventitia: Thin.

Artery: Aorta



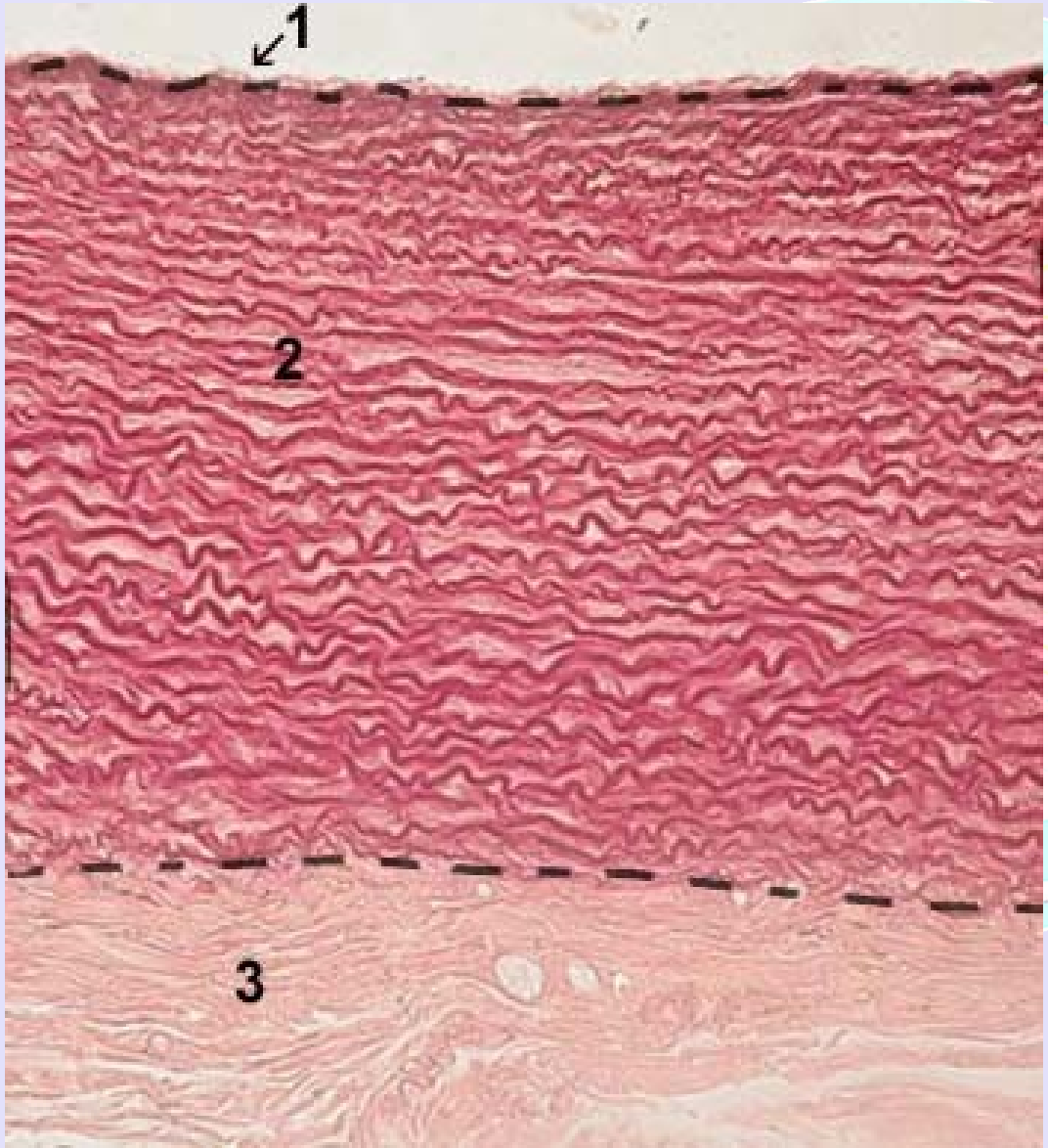
Muscular artery (left) & Elastic artery (right)



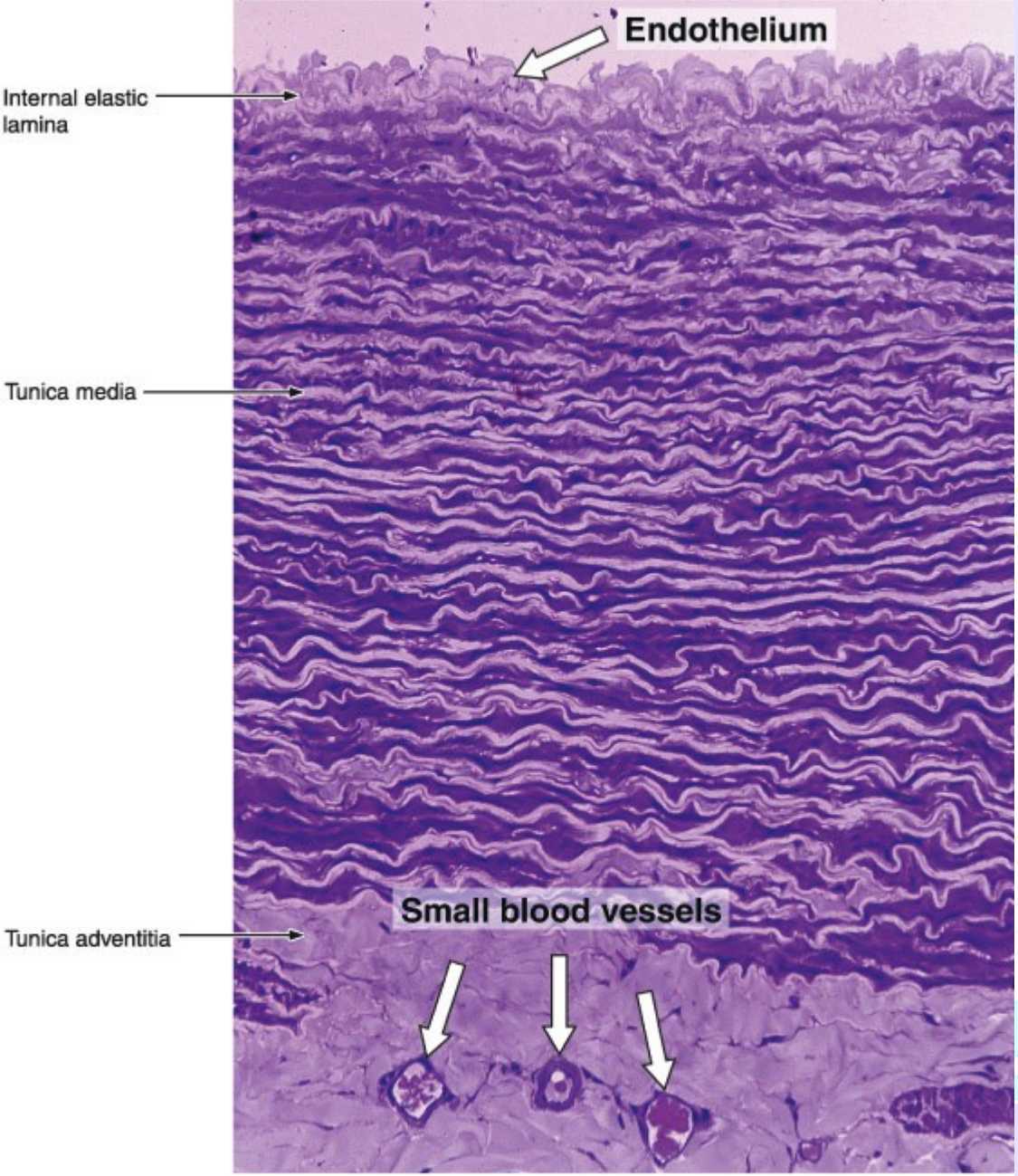
The tunica media of muscular artery contains predominantly smooth muscle, whereas the tunica media of an elastic artery is formed by layers of smooth muscle intercalated by elastic laminas. The adventitia and the outer part of media have small blood vessels (vasa vasorum) and elastic& collagenous fibers.

ELASTIC ARTERY (AORTA)

Stained with orsein



1 - tunica intima
2 - tunica media
3 - tunica externa



Source: Junqueira LC, Carneiro J: *Basic Histology: Text and Atlas*, 11th Edition: http://www.accessmedicine.com

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Transverse section showing part of a large elastic artery showing a well-developed tunica media containing several elastic laminas. Pararosaniline–toluidine blue (PT) stain. Medium magnification.

VEINS

"Capacitance vessels" because more than70%of total blood volume is in this portion of CVS at any time.

Structural difference B/W veins & arteries.

1.Caliber (Int.diameter) of vein is larger than same size artery, because veins have thinner walls due to scarcity of muscular and elastic components

2.Unlike arteries,small & medium sized veins have valves.

valves :-Consists of 2 semi lunar folds of tunica intima that project into the lumen.

They composed of elastic CT and are lined on both sides by endothelium.

They are specially numerous in veins of lower limbs.

COMPARISON 0F MEDIUM SIZE ARTERY & VEIN

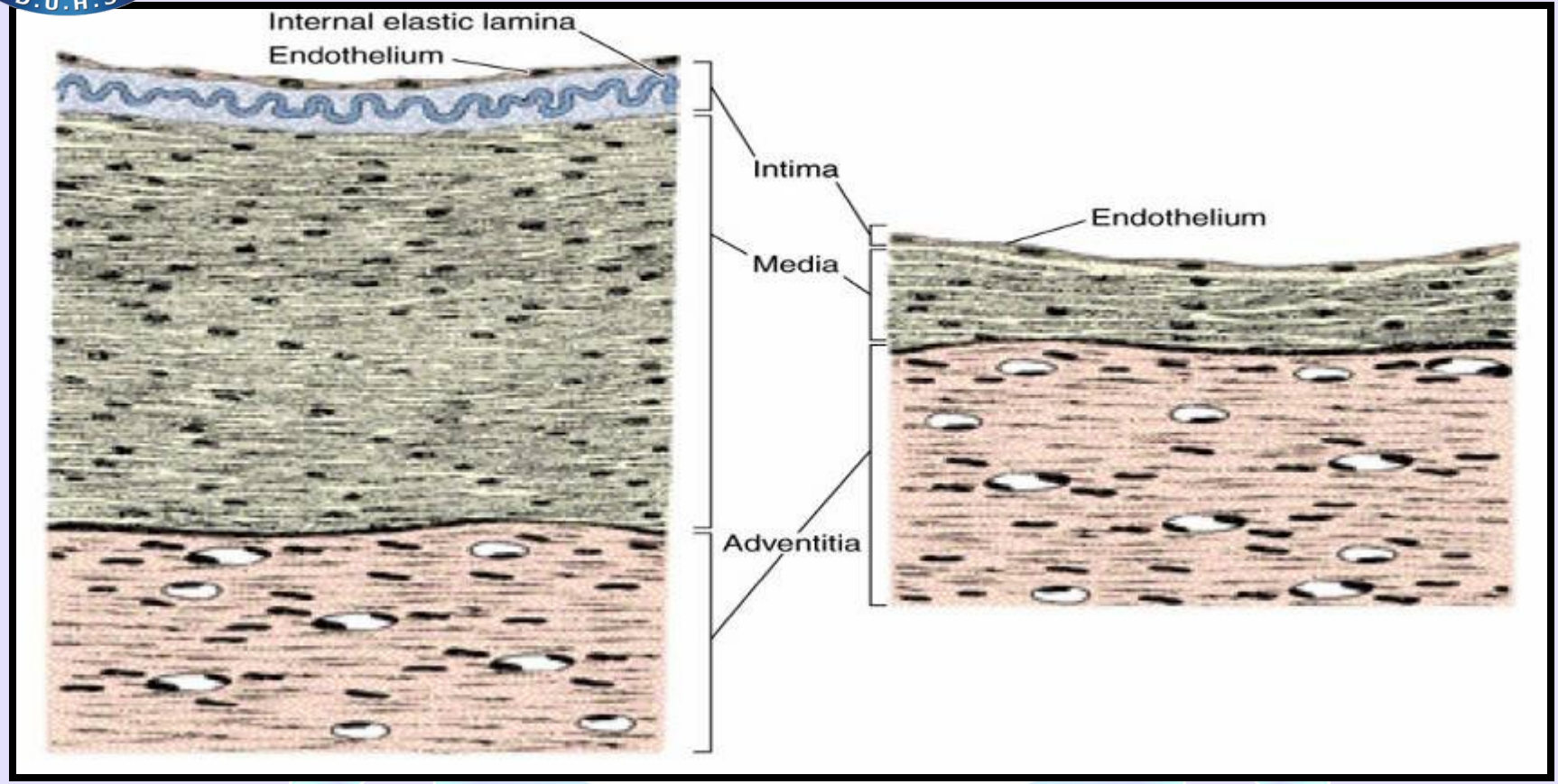


Diagram comparing the structure of a muscular artery (left) and accompanying vein (right).

Note that the tunica intima and tunica media are highly developed in the artery but not in the vein

A Comparison of a Typical Artery and a Typical Vein

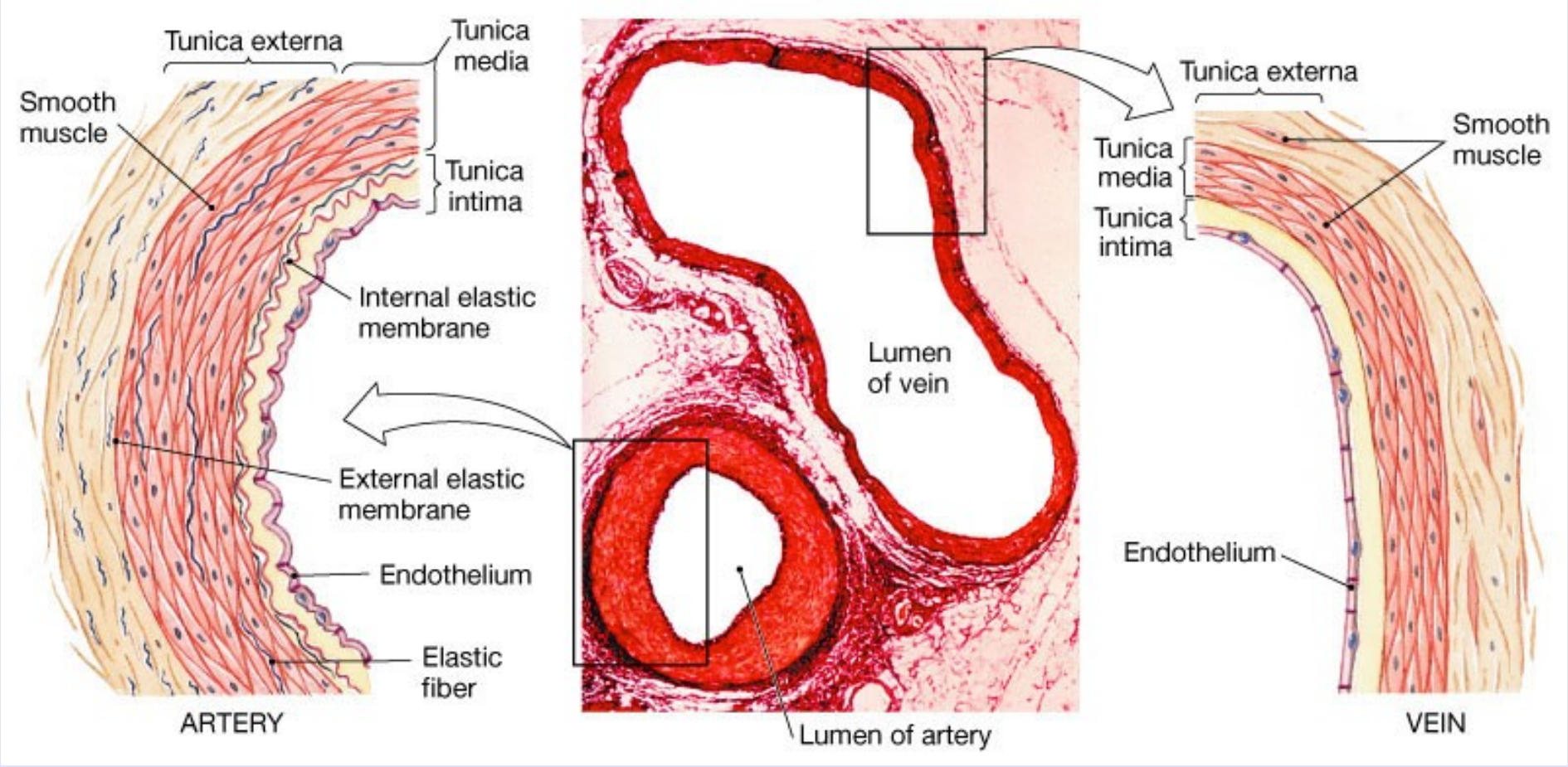
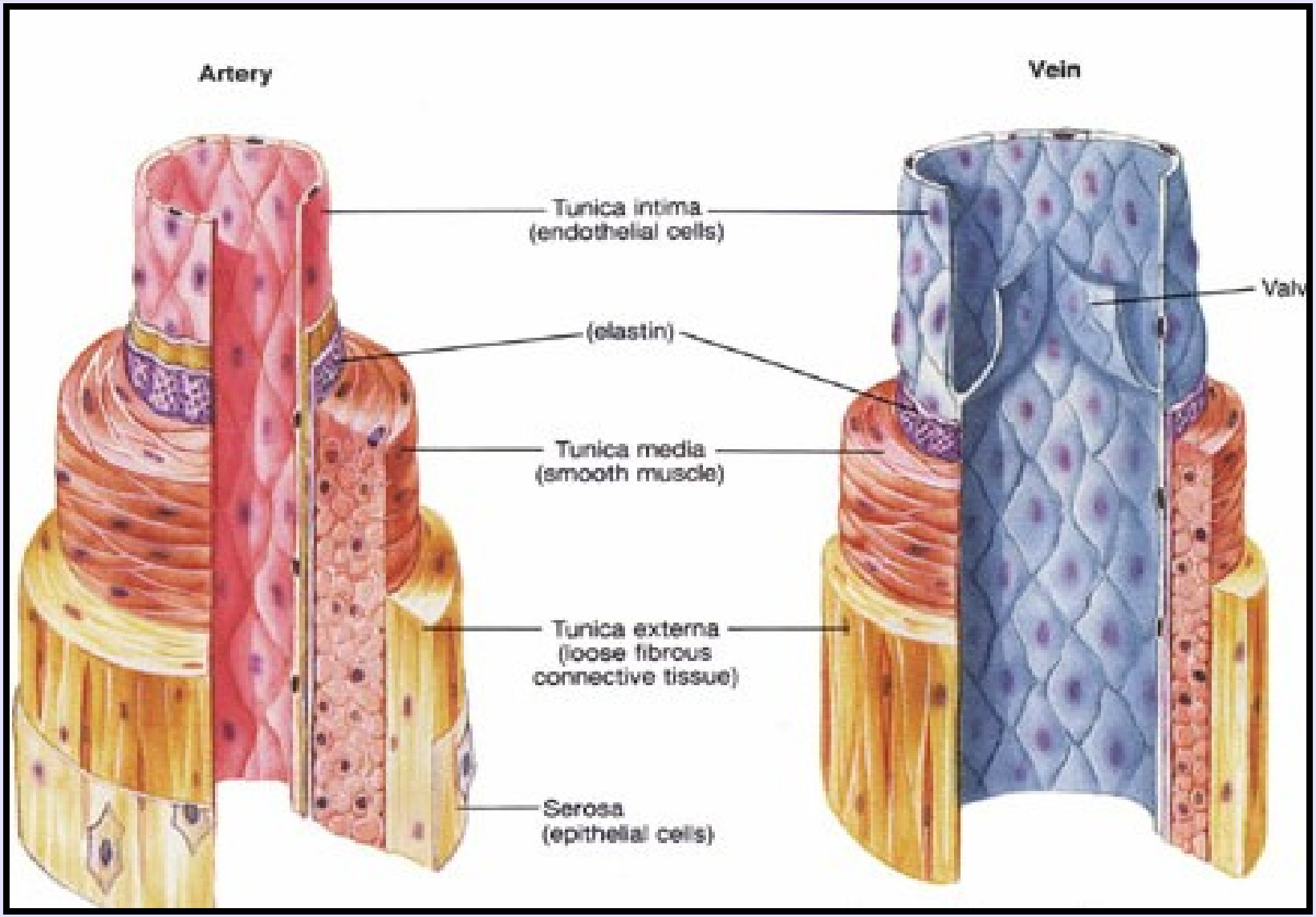


Figure 21.1

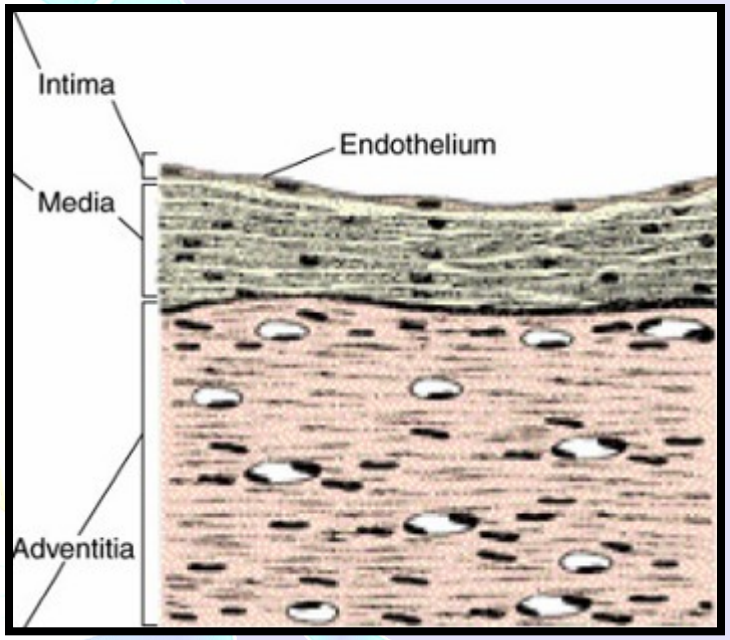
Comparison of artery and vein



VEINS

4 types, according to size.

1. **Venule.**
2. **Small veins.**
3. **Medium sized veins.**
4. **Large veins.**



VEINS *Cont.*

VENULE

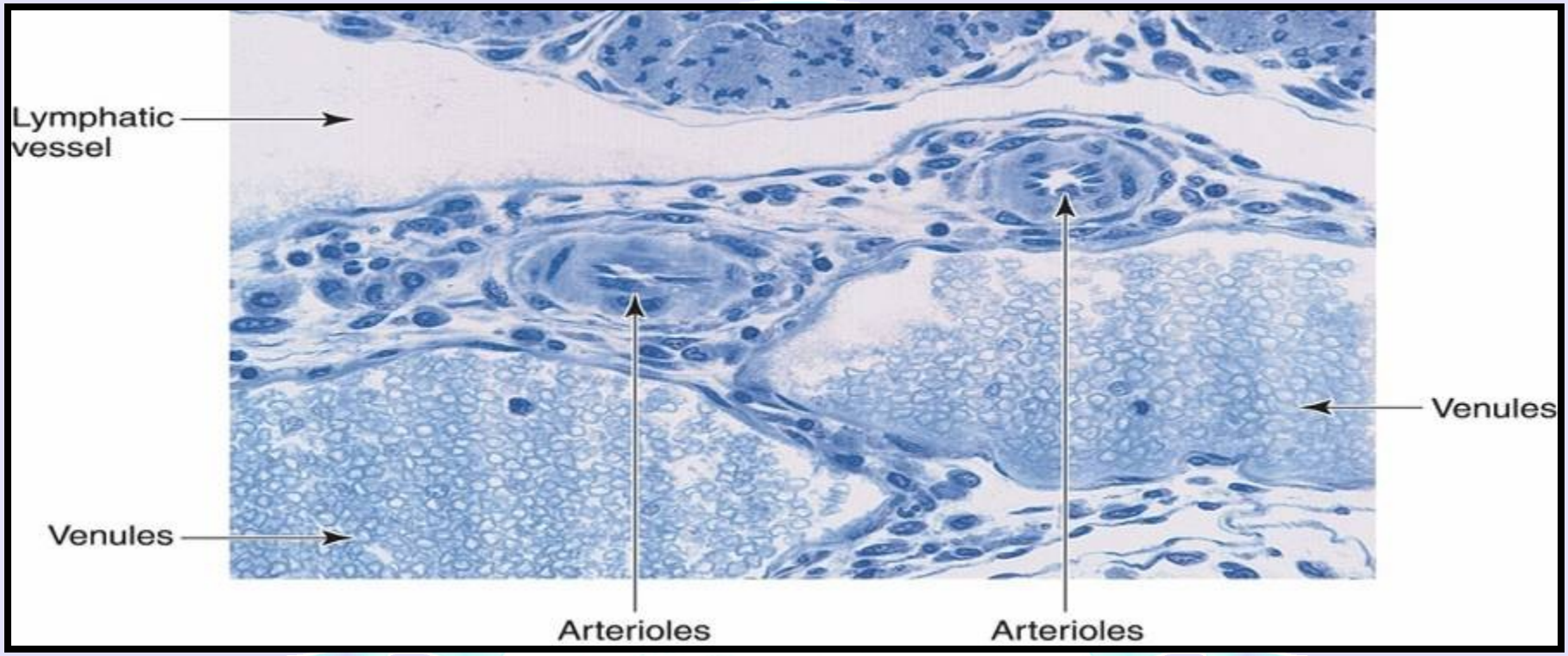
Diameter from 0.2 to 1mm.

T. intima: Only Endothelium

T. media: None to few smooth muscle fibers.

T. adventitia: Relatively Thickest, Consists of longitudinally oriented collagenous fibers.

Cross section of two venules and two small arterioles

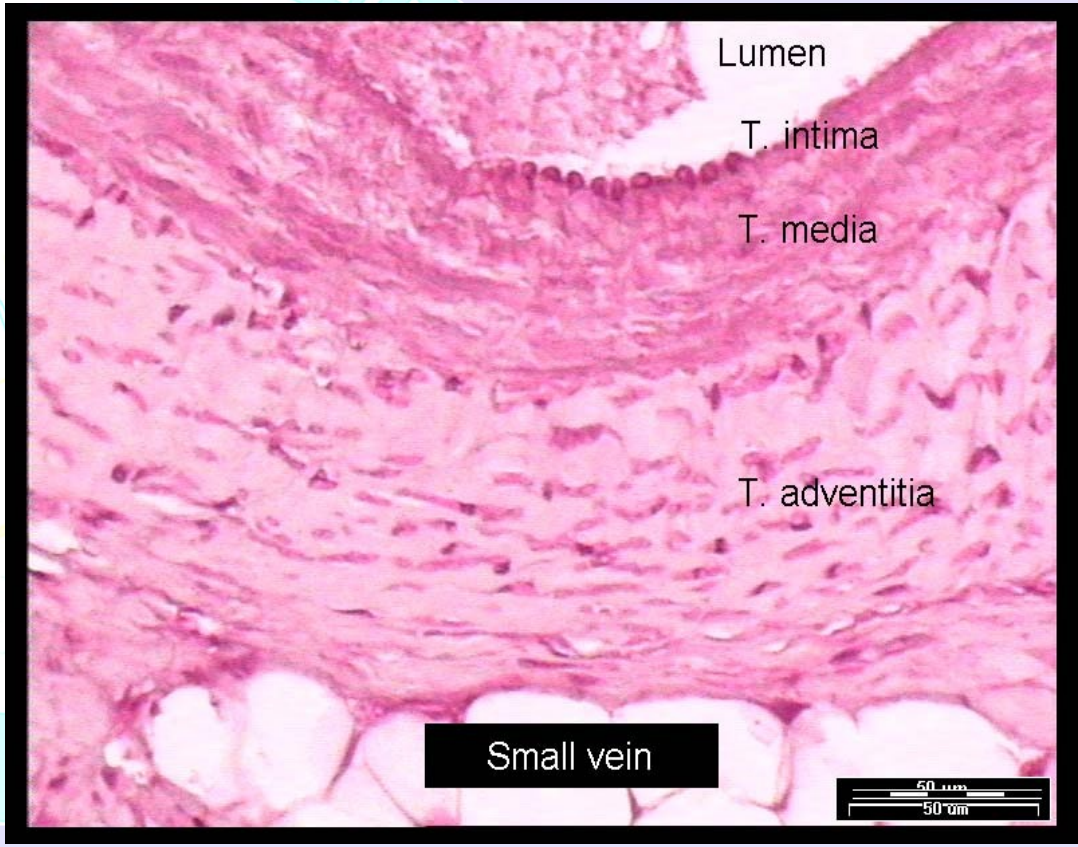


The walls of the arteries are thicker than the walls of the veins.

Note the cross sections of smooth muscle cells and the field of loose connective tissue that surrounds the vessels.

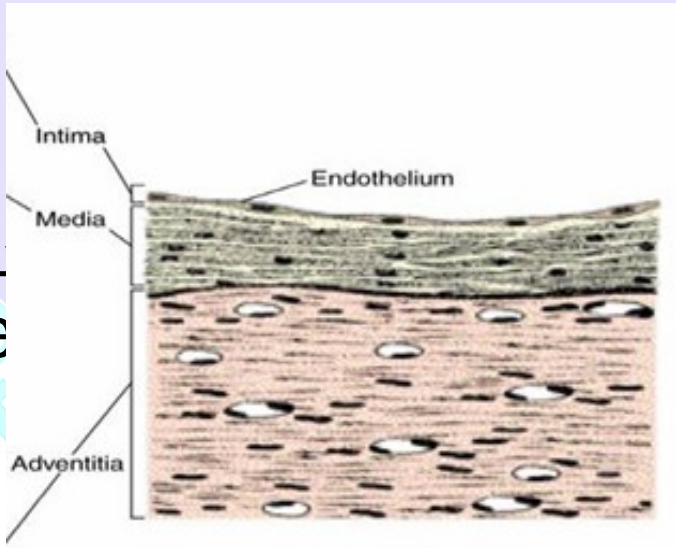
Small veins

- They have similar structure of venules, except, diameter (1 – 2 mm),
- few smooth layers in the tunica media and presence of internal valves.



VEINS *Cont.*

MEDIUM SIZED VEIN



Diameter from 2 to 9 mm. Example – Named veins.

T. intima:

Endothelium

Sub-endothelium: Thin

Internal elastic lamina: May be distinguish

T. media: Small bundles of smooth muscle fibers intermixed with reticular & delicate network of elastic fibers.

T. adventitia: Well developed. Forms bulk of vein's wall, composed of thick bundles of collagenous fibers, running in longitudinal direction.

VEINS *Cont.*

LARGE VEINS

Examples – superior & inferior vena cava & portal vein.

T. INTIMA

Endothelium

Sub-endothelium: Thin

Internal elastic lamina: May be distinguish

T. MEDIA Thin & poorly developed.

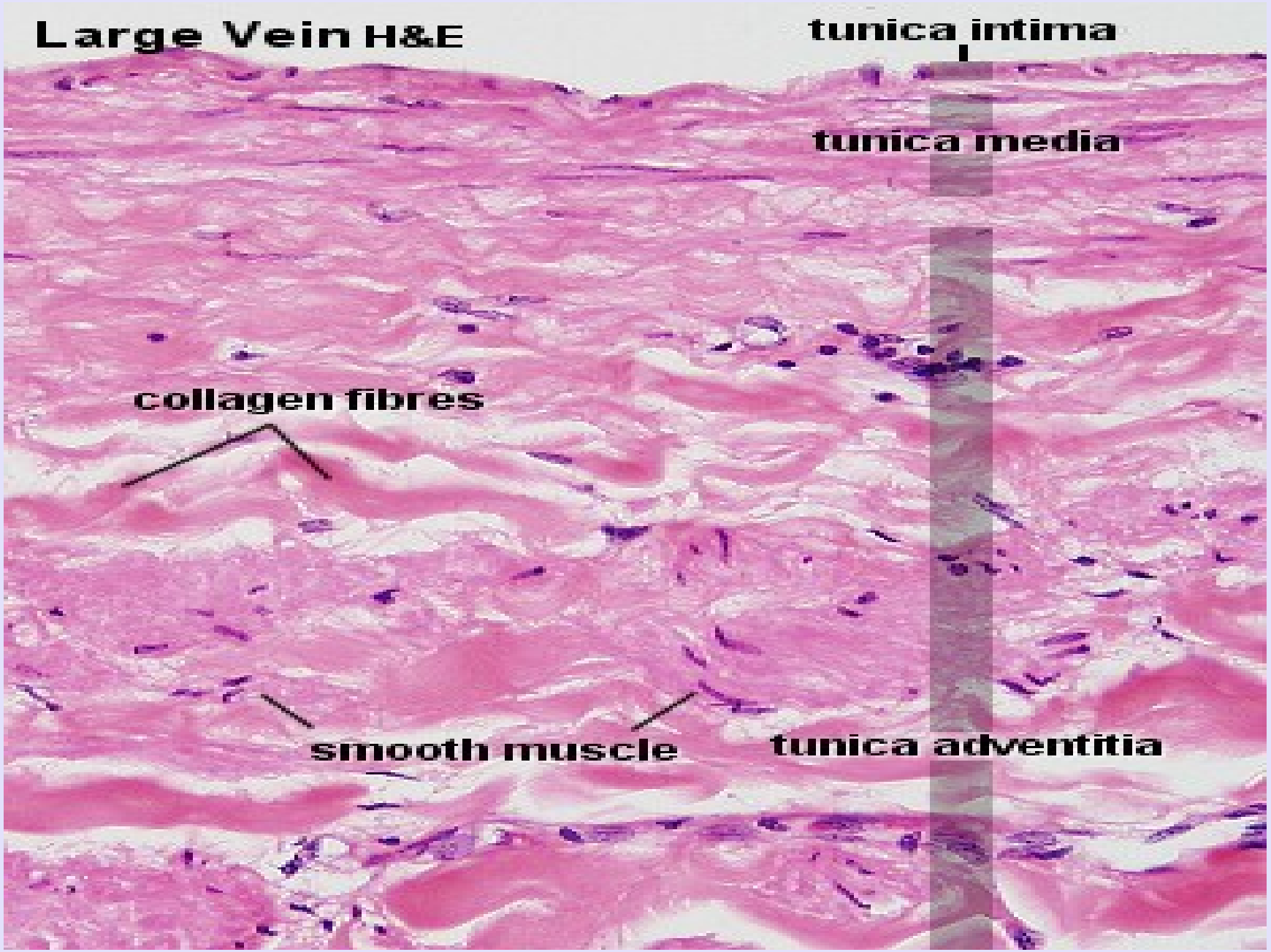
Mainly consists of collagenous fibers.

T. ADVENTITIA EXTREMELY THICKEST & has 3 ZONES

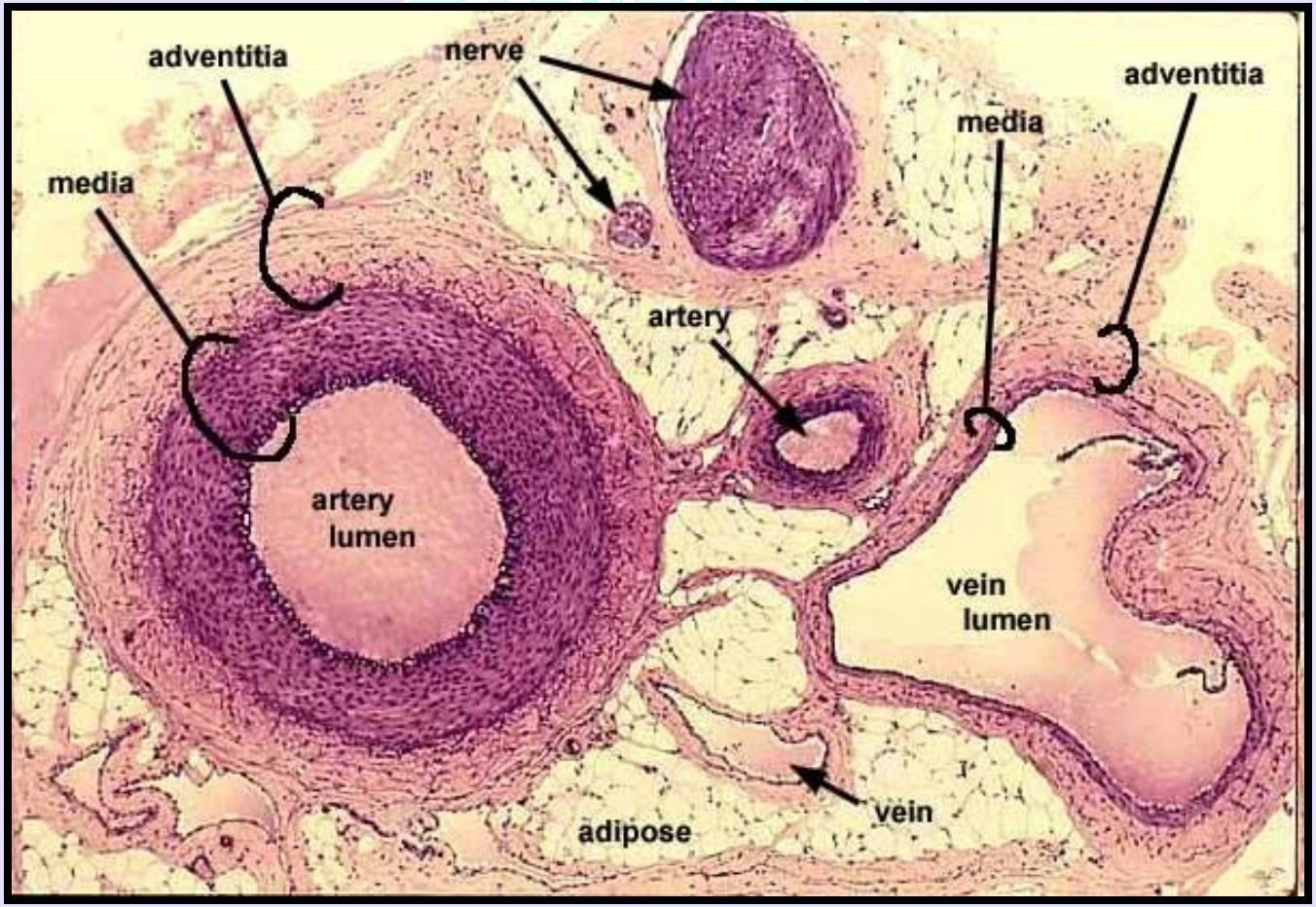
Inner most: Dense fibroelastic C.T. with coarse collagenous fibers, running spirally.

Middle: Longitudinally running smooth muscle fibers.

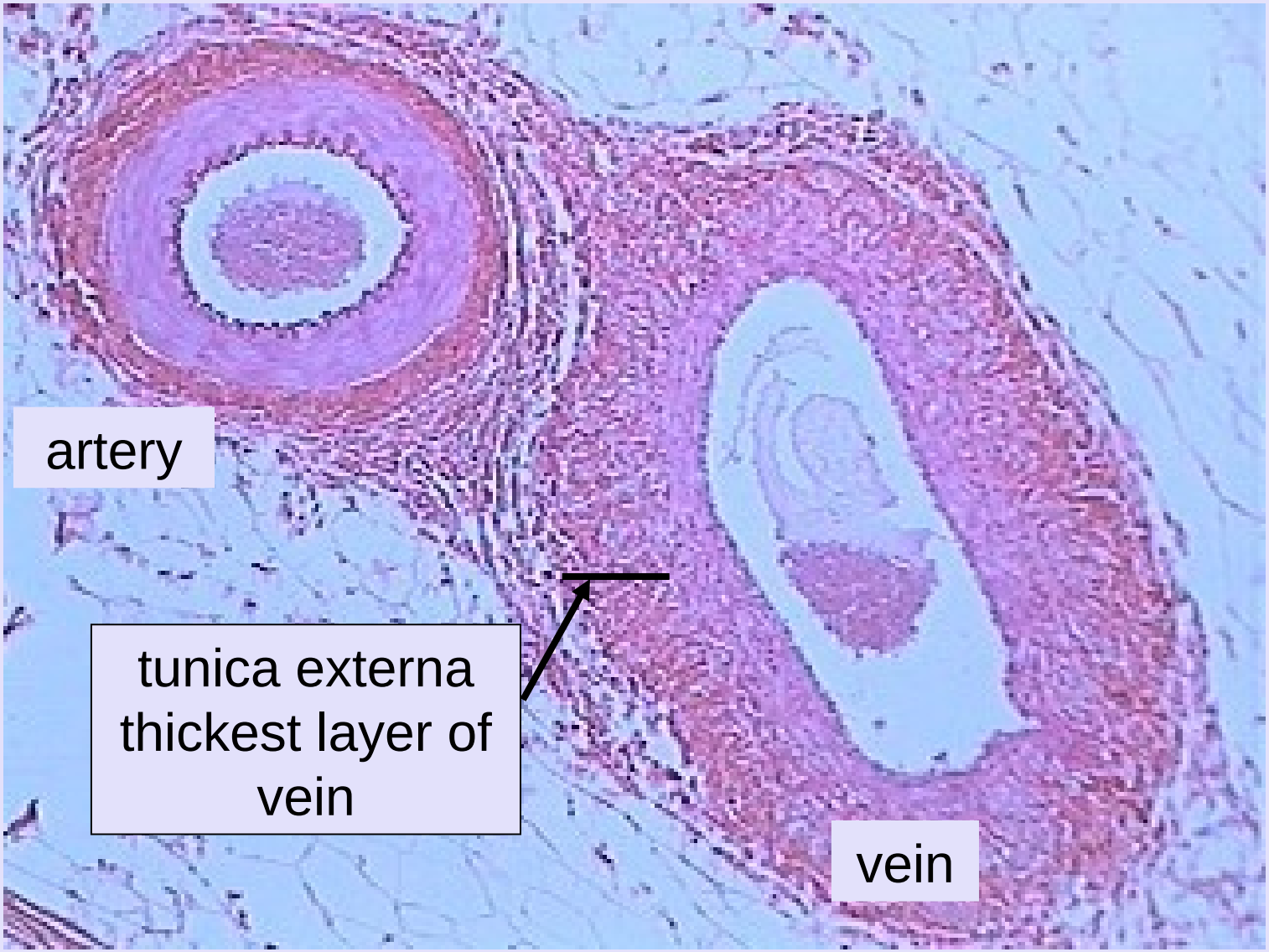
Outer: A network of coarse collagenous & elastic fibers.

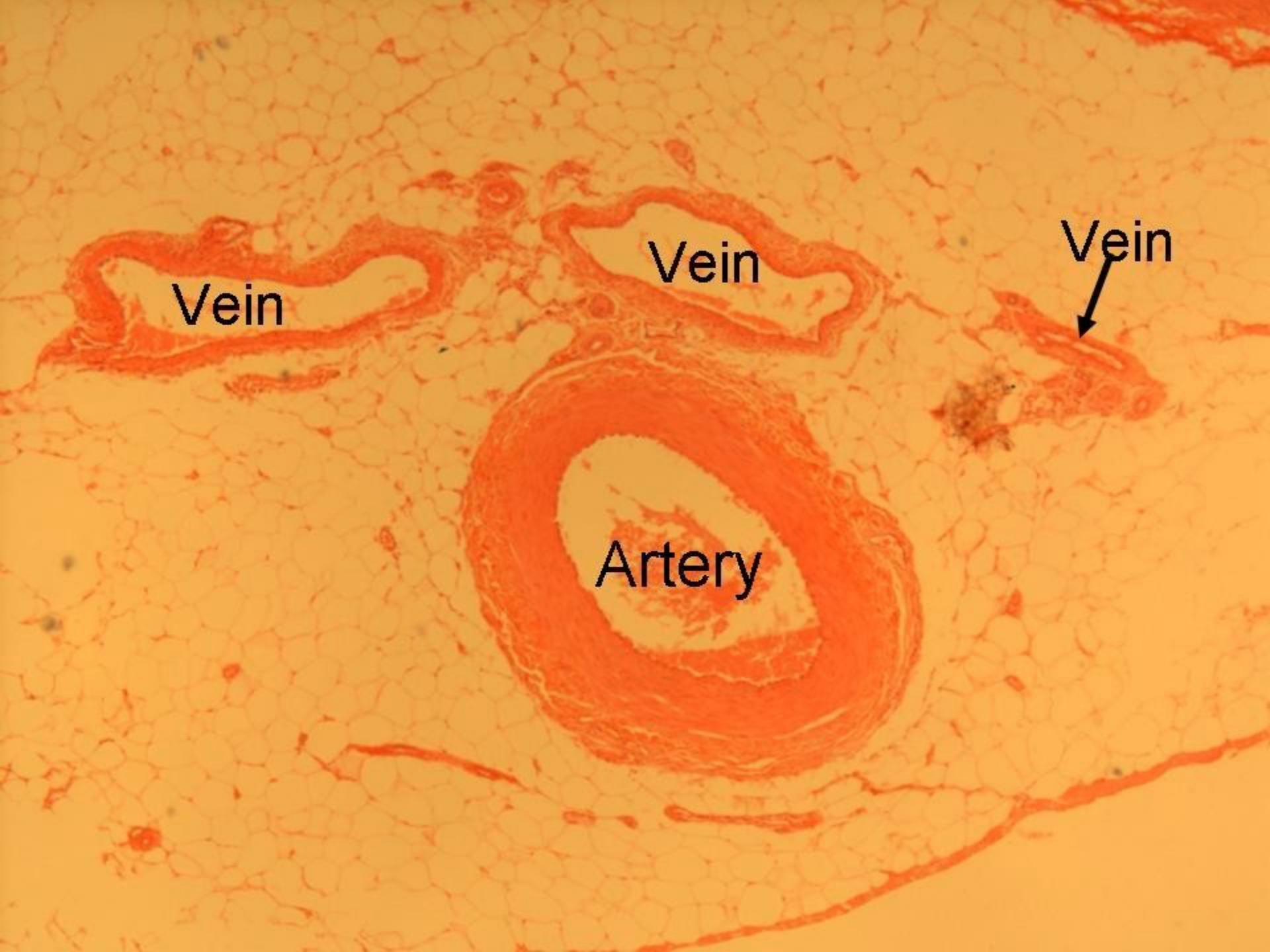

Large Vein H&E
tunica intima
tunica media
collagen fibres
smooth muscle
tunica adventitia

Artery, vein, nerve, elastin stain



Artery and Vein 100X




Vein
Vein
Vein
Artery

Comparison of artery, vein and capillary

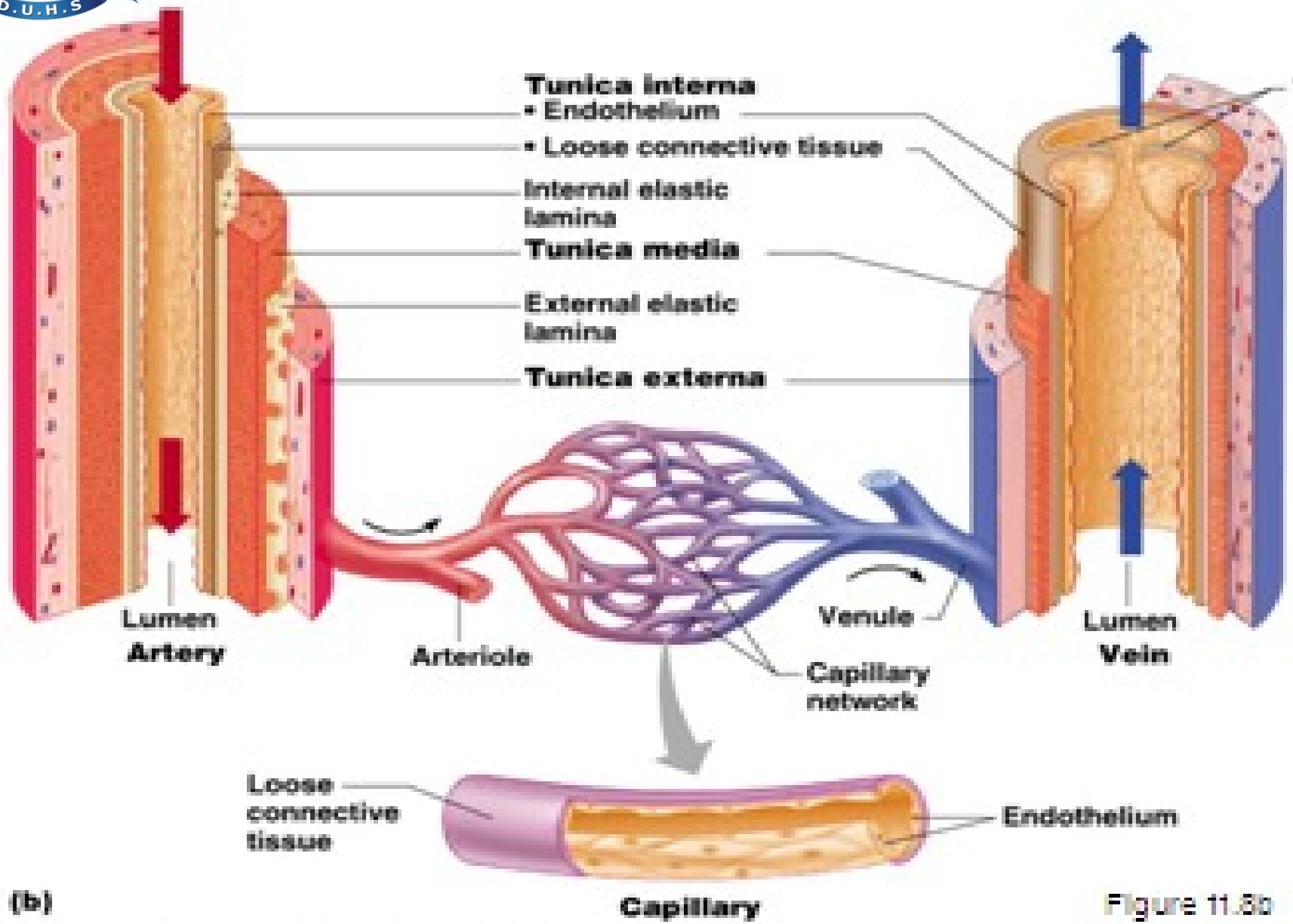


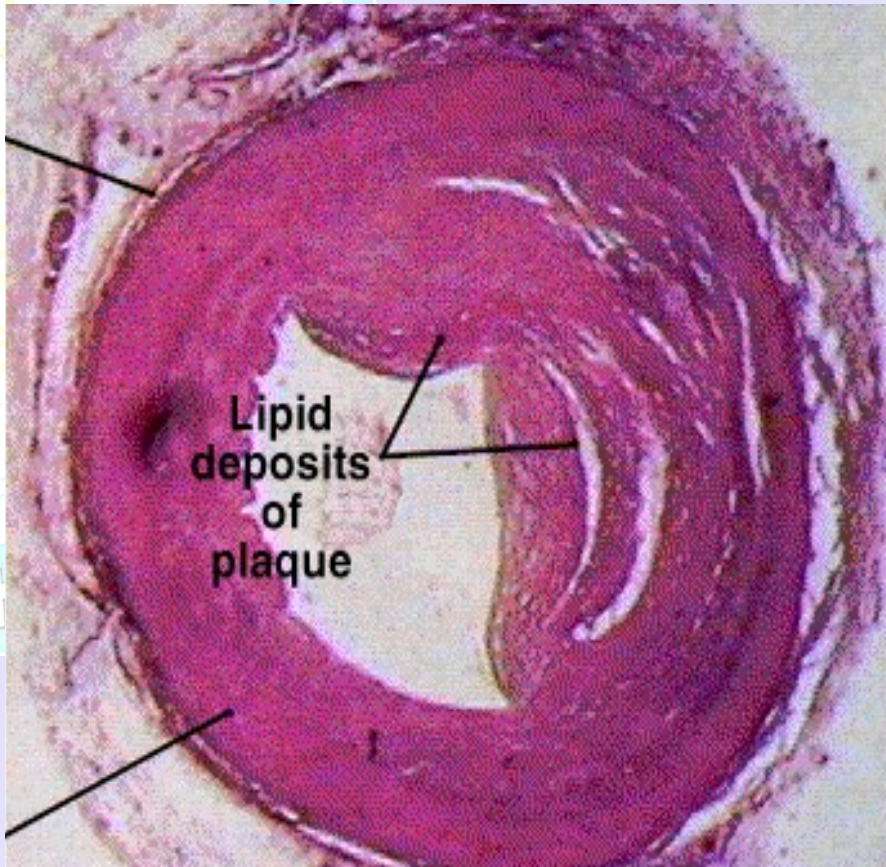
Figure 11.8b

Blood Vessel Pathologies

1. Aneurysms

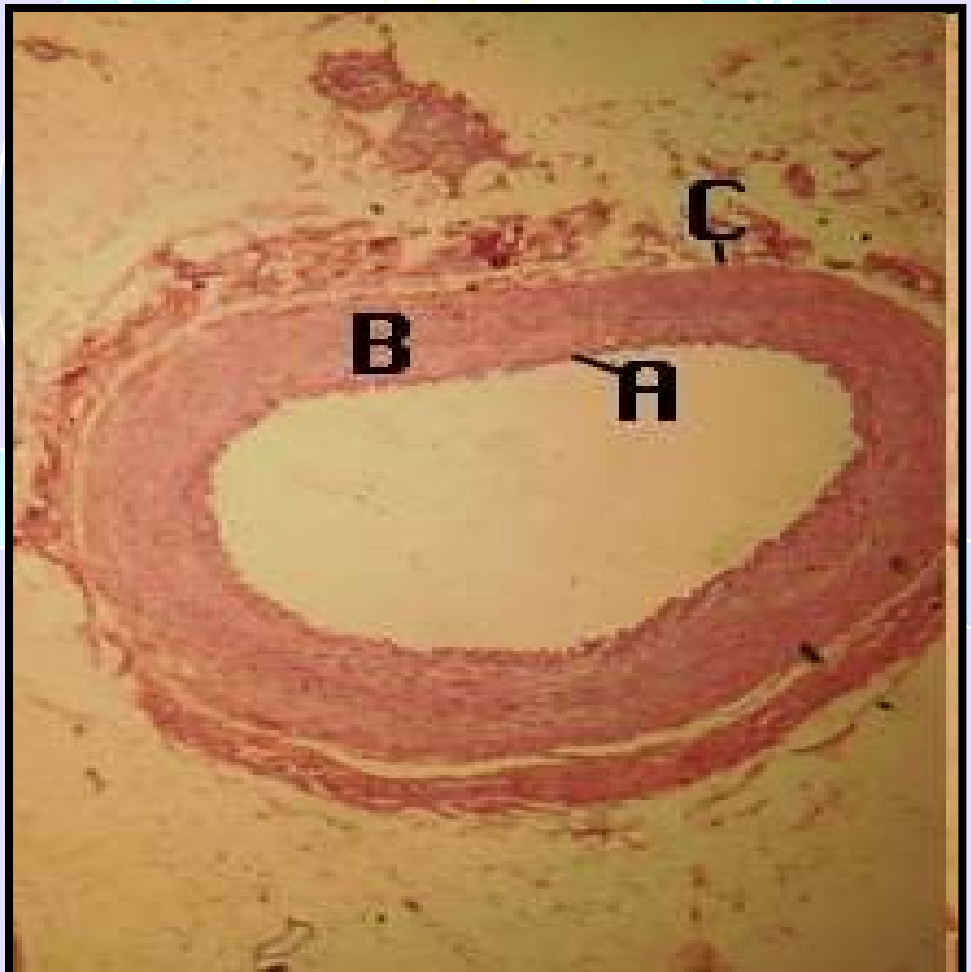
2. **Atherosclerosis = type of Arteriosclerosis)** thickening and toughening of arterial walls

Pathogenesis covered in Physiology



Self Assessment

Identify the vessel; name the layers marked as A,B and C



Self Assessment

Identify the vessel; name the layers marked as A,B and C



REFERENCE

- **MEDICAL HISTOLOGY BY LAIQUE HUSSAIN**

  5TH EDITION

THANK YOU